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Election-year stimuli and economic performance: Evidence from a macroeconometric model of the Philippines

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Abstract

We evaluated the transmission of election shocks in the Philippine economy using an augmented macroeconometric model that integrates political business cycle (PBC) dynamics into the country's macroeconomic framework. Building upon the model developed by Debuque-Gonzales and Corpus (2023, 2024), quarterly data from 2002 to 2023 were utilized to simulate the effects of election-induced fiscal and private sector behavior on key macroeconomic variables, namely private consumption, employment, investment, and government consumption. Results reveal that election years generate short-term, demand-driven expansions, fueled by heightened government spending, campaign activities, and temporary job creation. However, these effects are transitory, with economic activity reverting near baseline levels post-election as fiscal impulses fade. Findings align with established literature on political budget cycles, confirming that election-driven growth is cyclical rather than structural, and may induce inefficiencies in expenditure allocation and fiscal discipline. The study highlights the need for institutional reforms, fiscal transparency, and counter-cyclical policies to mitigate volatility and promote long-term stability. Finally, limitations related to model stability, pandemic disruptions, and evolving post-COVID economic structures suggest avenues for recalibrating and rewriting the macroeconometric model for future applications.

Keywords: election shocks; macroeconometric modeling, political business cycles

JEL Classification: C51, E62

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List of Abbreviations

ADB	Asian Development Bank
ADF	Augmented Dickey–Fuller
AIC	Akaike Information Criterion
ARDL	Autoregressive Distributed Lag
BIR	Bureau of Internal Revenue
BOC	Bureau of Customs
BSP	Bangko Sentral ng Pilipinas
COA	Commission on Audit
COMELEC	Commission on Elections
COVID-19	Coronavirus Disease 2019
CPI	Consumer Price Index
CPBRD	Congressional Policy and Budget Research Department
CUSUM	Cumulative Sum
DBM	Department of Budget and Management
DEPDev	Department of Economy, Planning, and Development
DLSU	De La Salle University
DOH	Department of Health
DSGE	Dynamic Stochastic General Equilibrium
ECM	Error Correction Model
FDI	Foreign Direct Investment
FRB	Federal Reserve Bank
GDP	Gross Domestic Product
GE	General Equilibrium
GFC	Global Financial Crisis
GMM	Generalized Method of Moments
HANK	Heterogeneous Agent New Keynesian
IMF	International Monetary Fund
IS-LM	Investment-Saving and Liquidity Preference-Money Supply
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MEM	Macroeconometric Model
NBER	National Bureau of Economic Research
NEDA	National Economic and Development Authority
NG	National Government
OECD	Organisation for Economic Co-operation and Development
OLG	Overlapping Generations
PBC	Political Business Cycle(s)
PIDS	Philippine Institute for Development Studies
PPP	Public-Private Partnership
PSA	Philippine Statistics Authority
RBA	Reserve Bank of Australia
SVAR	Structural Vector Autoregression
VAR	Vector Autoregression
ADB	Asian Development Bank

Election-Year Stimuli and Economic Performance: Evidence from a Macroeconometric Model of the Philippines

Mark Gerald C. Ruiz¹, Ramona Maria L. Miral², and John Paolo R. Rivera³

1. Introduction

The Philippine economy has historically shown fluctuations during election years, distinct from non-election periods, driven by increased government expenditures, heightened consumer and business activity, and shifts in investor confidence (Ochave 2025; Habito 2013; Landingin 2010). Elections, being periodic events when political and economic forces interact, impact macroeconomic performance through economic shocks that are transmitted to changes in fiscal policy, public infrastructure spending, capital flows, and overall market sentiment (de Haan et al. 2023). However, this growth can be unsustainable if it is not driven by structural economic improvements (Curtis 2023). Thus, such periods also introduce uncertainty, affecting inflationary trends, foreign direct investment (FDI) inflows, and long-term fiscal sustainability (Azzimonti 2024; Goodell et al. 2020; Gupta et al 2015).

The year 2025, being an election year in the Philippines (i.e., midterm elections⁴), differs from non-election years. On one hand, election years often see an uptick in gross domestic product (GDP) growth due to increased fiscal expenditures, particularly on infrastructure projects (i.e., to demonstrate accomplishments and gain voter support, governments often accelerate infrastructure programs in the months leading up to an election), social assistance programs (i.e., increased funding for social programs, subsidies, and cash transfers is common to bolster political goodwill), and election-related administrative spending (i.e., budget for electoral processes, including voter registration, election security, and logistics, contributes to government spending growth) (Das et al. 2025; Olano 2019). Also, the government tends to frontload disbursements in the run-up to elections, driving short-term demand in the economy (Frieden 2020). Succeeding, post-election years often see fiscal tightening as governments attempt to curb deficits and adhere to fiscal discipline, leading to a slowdown in public investment (Tannous 2024).

Consequently, the increased liquidity in the economy during election years, stemming from government expenditures and campaign-related spending, can contribute to higher inflation (Kladakis and Skouralis 2024). The surge in money supply, combined with increased consumer demand, can lead to price increases, particularly in food, services, and transportation. Moreover, if the government resorts to deficit spending to finance election-related expenditures, inflationary pressures may persist beyond the election period. While GDP growth may temporarily rise, election-related uncertainty often dampens investor confidence (Azzimonti 2024; Goodell et al. 2020; Gupta et al 2015). FDI inflows tend to slow during election years as businesses and investors adopt a wait-and-see approach, assessing potential shifts in policy, regulatory frameworks, and political stability (Jahn and Stricker 2022). Political transitions can lead to concerns over changes in business conditions, tax regimes, and contract security, prompting investors to delay commitments or divert capital to more stable environments

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⁴ In the Philippine context, a midterm election refers to a national and local election held halfway through the six-year term of the incumbent president. It takes place three years after a presidential election and serves as a political barometer of the sitting administration's performance and public approval.

(Boyles 2022). The impact on FDI flows depends on the perceived credibility of institutions and the policy stance of incoming leadership (Kapas 2020).

On the other hand, election years also see significant uptick in private sector and household expenditures (Olano 2019). Political campaigns inject significant liquidity into the economy through salaries, advertising expenditures, and logistics, leading to higher consumer spending, particularly in services, retail, and transport. Likewise, some businesses increase spending during election years, particularly those linked to election-related industries such as media, advertising, and printing (Le et al. 2024). However, firms with long-term investment plans may delay major capital expenditures due to policy uncertainty and a wait-and-see behavior (Azzimonti 2024; Goodell et al. 2020; Gupta et al 2015).

Given the May 2025 Philippine midterm elections, our study is relevant in providing empirical insights on the extent of election-driven economic stimulation. While election-related government expenditures may create short-term growth, concerns remain about inflationary pressures, fiscal sustainability, and economic volatility post-election. Moreover, investor sentiment (both domestic and foreign) can be swayed by electoral uncertainty, impacting capital inflows and business decisions.

Given the abovementioned backdrop and the cyclical nature of these economic dynamics, we inquire on *how election-related shocks are transmitted through the Philippine macroeconomy?* In addressing this research question, we are guided by an overarching objective *to integrate election variables into a macroeconometric model for the Philippines*. Supporting this are the following specific objectives:

1. To estimate the magnitude of election-year stimulus on GDP growth and whether it leads to sustained economic benefits or short-term artificial growth;
2. To estimate the extent to which inflation is influenced by election-related liquidity surges, private sector expenditures, and government spending patterns;
3. To evaluate how election-induced uncertainty affects FDI inflows and business confidence; and
4. To generate policy recommendations on how the government can balance economic stimulus with long-term stability.

By evaluating the transmission of election shocks through a macroeconometric framework developed by Debuque-Gonzales and Corpus (2023), we contribute to evidence-based policymaking, helping stakeholders better understand the economic implications of election cycles. Also, by offering a structured analysis of these effects, we contribute to informed decision-making for policymakers, economists, and business leaders navigating the complexities of election-year economic shifts and in anticipating both short-term economic boosts and potential long-term distortions. Findings can serve as a basis for formulating strategies to mitigate economic risks while leveraging potential benefits during election periods.

2. Literature review

2.1. Historical evolution of macroeconometric models

The origins of macroeconometric modeling trace back to early macroeconomic thought on general equilibrium (GE) theory, which introduced aggregate demand as an analytical variable distinct from aggregate supply. Keynes (1936) challenged the classical notion of self-correcting markets by arguing that economies could settle into equilibrium with involuntary

unemployment, requiring government intervention to stimulate demand. Hicks (1937) formalized this interaction in the IS-LM (investment-saving and liquidity preference-money supply) framework, later integrated into the “neoclassical synthesis” that combined long-run neoclassical principles with short-run Keynesian dynamics (Samuelson 1948).

The post-war period saw the rise of large-scale econometric models, spurred by the Cowles Commission’s⁵ pioneering work in statistical estimation and model testing. A major turning point came with the Lucas Critique⁶ (Lucas 1976). This critique spurred the development of micro-founded models grounded in rational expectations, including the policy-ineffectiveness proposition of Sargent and Wallace (1975).

In subsequent decades, macroeconomic modeling evolved toward Dynamic Stochastic General Equilibrium (DSGE) frameworks that explicitly incorporate intertemporal optimization and stochastic shocks (Kydland and Prescott 1982; Smets and Wouters 2007). Later refinements introduced New Keynesian DSGE models that allow for nominal rigidities (Calvo 1983) and Heterogeneous Agent New Keynesian (HANK) models that account for distributional effects (Kaplan et al. 2018). Despite their sophistication, DSGE models faced criticism for failing to anticipate the 2008 Global Financial Crisis (Hendry and Muellbauer 2018; Stiglitz 2018). Contemporary work thus emphasizes more flexible and data-driven macroeconomic approaches integrating behavioral expectations, non-linearities, and institutional factors (Blanchard 2016; Guerrieri and Iacoviello 2017; Albuquerque et al. 2025).

2.2. Existing macroeconomic models in the Philippines

The Philippines’ experience with macroeconomic modeling has evolved alongside global developments. Early models were developed by academic institutions⁷, multilateral organizations⁸, and government agencies such as the Philippine Institute for Development Studies (PIDS), the Department of Economy, Planning and Development (DEPDev⁹), and the Bangko Sentral ng Pilipinas (BSP¹⁰). These models, often medium-scale and demand-driven, were designed to inform fiscal and monetary policy by simulating relationships among aggregate output, consumption, investment, and inflation (Debuque-Gonzales and Corpus 2023, 2024). For a historical background of earlier macroeconomic models in the Philippines, Debuque-Gonzales and Corpus (2023, 2024) provided a detailed discussion from the annual macroeconomic model (MEM) by Constantino and Yap (1988), Constantino et

⁵ Focuses on linking economic theory to mathematics and statistics; its advances in economics involved the creation and integration of GE theory and econometrics.

⁶ A theory in macroeconomics that criticizes the use of past data to predict how new economic policies will affect the economy. It argued that people’s expectations and behavior change in response to new economic policies. Therefore, using historical data to predict the effects of new policies is not reliable. Moreover, it is a fundamental criticism of empirical economics, which questions its ability to model, test, or predict the economy (Stanley 2000).

⁷ For instance, De La Salle University (DLSU) has developed multiple macroeconomic models of the Philippine economy (i.e., ANIMO model, Quarterly model, Simultaneous equation system, OLG model). These models are used to forecast the economy and help inform policy decisions. For more information, see <https://www.dlsu-aki.com/research-programs-and-projects.html>. Also, Rodriguez and Briones (2002) built the quarterly Ateneo Macroeconomic and Forecasting Model (AMFM) based on the short-run version of the Murphy model of Australia.

⁸ The Asian Development Bank (ADB) developed macroeconomic models of select ADB member economies for forecasting and policy simulation. The model designed for the Philippines (Cagas et al. 2006; Ducanes et al. 2005) paid special attention to the government block of the model to enable fiscal simulations. See <https://www.adb.org/sites/default/files/publication/28191/wp062.pdf>.

⁹ Formerly known as the National Economic and Development Authority (NEDA); the Philippine government agency responsible for national economic planning, policy coordination, and monitoring to ensure sustainable development; also oversees the approval of large projects, trade policies, and the efficient use of land and natural resources, working to link development planning with the national budget. Website: <https://depdev.gov.ph/>.

¹⁰ The Policy Analysis Model for the Philippines (PAMPh) is a model used by the BSP to analyze the economy and guide monetary policy. It is used as the main model for medium-term forecasting and policy analysis (Alarcon et al. 2020). See <https://www.bsp.gov.ph/Sites/researchsite/Publications/BSP-Working-PaperSeries/WPS202012.pdf>.

al. (1980), Reyes and Yap (1993), Yap (2000) to more recent macroeconometric and forecasting model by Rodrigues and Briones (2002), structural MEM by Ducanes et al. (2005) and Cagas et al. (2006), quarterly macroeconometric model of Bautista et al. (2009), annual MEM by Reyes et al. (2020),

Debuque-Gonzales and Corpus (2023, 2024) made a significant step toward systematizing this framework using robust econometric specifications. These models align with international standards by employing simultaneous equations, error-correction mechanisms, and cointegration analysis to capture both short- and long-run dynamics. However, political and institutional shocks, particularly election-induced fluctuations, remain underexplored within these models. We build on this gap by integrating election shocks as exogenous or structural disturbances that influence fiscal behavior, investor confidence, and macroeconomic outcomes.

2.3. Economic impacts of from election activities

Substantial scholarly literature has explored the two-way relationship between elections and the economy. On one hand, macroeconomic conditions affect voting behavior, known as economic voting, where voters reward or punish incumbents based on perceived economic performance (Bello 2021; Guntermann et al. 2021; Leigh 2004; Alvarez et al. 1999; Fiorina 1978).

Conversely, elections influence the economy through various channels. The political business cycle literature suggests that incumbents manipulate fiscal or monetary tools to enhance reelection prospects (Schultz 1995; Rogoff and Sibert 1988). Empirical studies show that election years often coincide with surges in public expenditures, shifts in credit conditions, or changes in money supply (Kolios, 2019; Peters, 2010). Recent cross-country evidence from Nguyen and Tran (2023) confirmed that incumbents in 91 emerging and developing economies expanded government spending before and during elections, then contracted it afterward.

Sector-specific analyses corroborate these macro-level findings. Broni et al. (2019) observed higher bank returns during Ghanaian election years as citizens increased deposits amid political uncertainty, while Tabash et al. (2024) found that the 2018 Pakistan general election positively affected stock market performance. Despite this broad literature, there remains limited empirical research assessing the multi-sectoral macroeconomic impact of elections in the Philippine context, creating space for our contribution.

2.4. Political business cycles

The concept of political business cycles (PBCs) provides the theoretical foundation for analyzing election shocks. PBC models posit that elected officials, seeking reelection, manipulate fiscal and monetary levers to generate temporary economic upswings before elections (Schultz 1995; Rogoff and Sibert 1988; Nordhaus 1975). Subsequent empirical work quantified these effects such as that of Coulombe (2021) who identified ideological and electoral drivers of fiscal expansions in OECD countries, as well as Cipullo and Reslow (2022) who documented politically motivated GDP forecast optimism in advanced economies.

These findings suggest that election shocks transmit through multiple macroeconomic channels (i.e., government spending, inflation expectations, investor sentiment), all of which can be modeled within a macroeconometric framework to understand cyclical fluctuations in output and employment.

2.4. Econometric approaches to modeling election shocks

Modeling election shocks requires distinguishing between anticipated and unanticipated components (Adams and Barrett 2023; Mertens and Ravn 2012). Anticipated shocks are often captured using dummy variables representing election periods to model systematic policy changes, particularly fiscal expansions (Ivanovic et al. 2023; Van Dalen and Swank 1996). Meanwhile, unanticipated shocks, such as surprise election outcomes or abrupt policy shifts, are typically modeled using Structural Vector Autoregression (SVAR) frameworks, which isolate the dynamic responses of key variables to unexpected shocks (Gambetti 2021; Hoke 2019). These econometric techniques are essential in quantifying how election cycles affect aggregate demand, investment, and inflation through different transmission mechanisms.

2.5. Model enhancements

Integrating political and institutional variables into macroeconometric models enhances both model calibration and policy design. From a modeling perspective, the inclusion of political variables allows for more precise calibration, improving the reliability of simulations and forecasts (Martinoli et al. 2022). From a policy standpoint, understanding election-driven cycles supports the formulation of counter-cyclical fiscal policies to offset inflationary pressures or post-election slowdowns (Jalles et al. 2023; IMF 2023). This integrated approach bridges empirical modeling with practical fiscal governance, making macroeconometric tools more responsive to real-world political dynamics.

2.6. Research gap

Despite an extensive body of scholarship on PBCs and macroeconomic modeling, there remains a dearth of studies that systematically examines how election shocks transmit through the Philippine economy within an econometric framework. Existing studies, both local and international, tend to focus on short-term fiscal dynamics, such as pre-election increases in government disbursements or post-election fiscal adjustments, or on sector-specific impacts like financial markets, inflation, or employment. These fragmented analyses, while valuable, fall short of capturing the economy-wide propagation mechanisms through which election-related disturbances affect aggregate output, consumption, investment, and external accounts.

In the Philippines, most empirical work has emphasized descriptive or correlational approaches rather than model-based simulations that integrate behavioral and structural relationships among macroeconomic variables. It lacks a comprehensive, empirically estimated framework that can quantify how political cycles, through fiscal impulses, policy uncertainty, or investor sentiment, interact with the broader macroeconomic system. This limits policymakers' ability to distinguish between temporary election-induced booms and sustainable growth drivers.

Our study addresses this analytical and methodological gap by developing an augmented macroeconometric model for the Philippines, building from the work of Debuque-Gonzales and Corpus (2023, 2024), that explicitly incorporates election shocks. This enhancement allows for a more rigorous understanding of the timing, magnitude, and persistence of election-related effects on macroeconomic variables. By embedding electoral cycles as exogenous shocks within the structural equations of the model, we can simulate how fiscal behavior, private investment, and external balances respond under different election scenarios. Hence, we contribute not only to the empirical literature on political business cycles but also to macroeconomic policy design, providing evidence-based insights into how political events

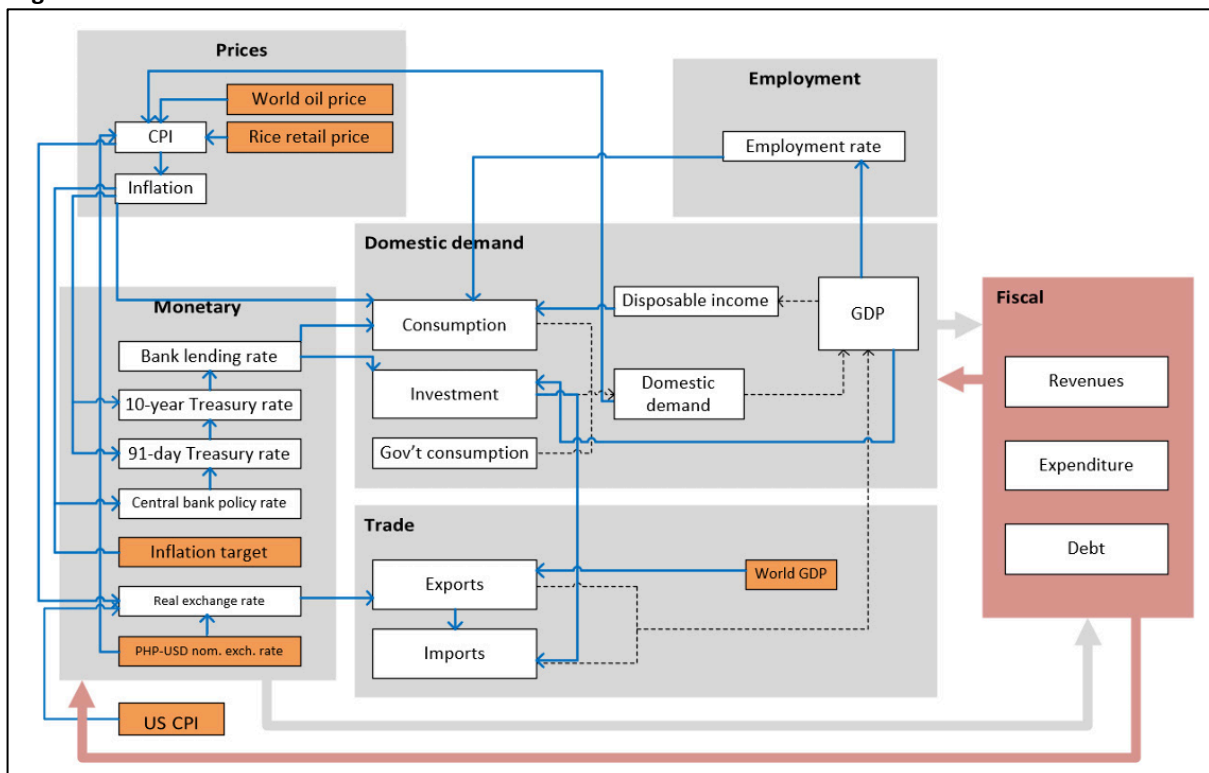
shape cyclical fluctuations, fiscal sustainability, and long-term growth trajectories. It also extends the capability of existing Philippine macroeconomic frameworks from merely describing economic trends to explaining and forecasting politically driven economic dynamics. This is a critical advancement for planning and fiscal management in the country.

3. Methodology

3.1. Conceptual framework

We build on the framework and model developed by Debuque-Gonzales and Corpus (2023, 2024) anchored on traditional macroeconomic models instead of micro-founded macroeconomic models in response to the Lucas Critique. We continue to align with an empirical approach to analyzing the transmission mechanism of election shocks to the macroeconomy following Hendry (2020) citing the continued use of econometric approach to macroeconomic analysis like the Federal Reserve Bank (FRB), the Norges Bank, the Reserve Bank of Australia (RBA), the Bank of Canada, and the European Central Bank employing non-DSGE models. Debuque-Gonzales and Corpus (2023, 2024) comprehensively reviewed the justifications for the continued preference for empirical approaches. By integrating election shocks into the macroeconomic model, we quantify how election cycles influence key macroeconomic indicators as seen in Figure 1.

Figure 1. Model structure



Note: Orange boxes denote the exogenous variables in the model. Solid blue lines represent behavioral relationships, while broken lines represent identities.

Source: Debuque-Gonzales and Corpus (2023, p. 21); Debuque-Gonzales and Corpus (2024, p. 6).

3.2. Operational framework

Following Debuque-Gonzales and Corpus (2023, 2024), we also adapted a pragmatic approach, where the objective was to build a policy model guided by economic theory but can fit data

reasonably well. We emphasized on usability, tractability, and ease of maintenance, apart from model validity and robustness. Table 1 shows the behavioral equations representing the macroeconomic variables in Figure 1.

Table 1. Model equations and variables

Equations	Variables
Domestic demand	C = Household consumption
$\log C_t = f(\log(YD_t), emp_t, r_t^{bl} - \pi_t^e, \pi_t^e)$	CPI = Consumer Price Index
$\log(I_t) = f(\log(Y_t), \Delta(r_t^{bl} - \pi_t^e), \pi_t^e)$	CPI ^{US} = US Consumer Price Index
$\log G_t = f(XP_t^{PR})$	D = National Government (NG) debt (nominal)
$\log M_t = f(I_t, X_t)$	D ^D = Domestic NG debt (nominal)
$Y_t \equiv C_t + I_t + G_t + X_t - M_t$	D ^F = Foreign NG debt (nominal)
$YD_t \equiv Y_t - RV_t^{TXBIR}$	emp = Employment rate
$\log(P_t^Y) = f(\log(CPI_t))$	G = Government consumption
$Y_t^N \equiv P_t^Y Y_t$	I = Investment
Trade block	M = Imports
$\log(X_t) = f(\log(Y_t^{WORLD}), \log(xrr_t))$	NX = Net exports
$\log(M_t) = f(\log(I_t), \log(X_t))$	p ^{oil} = World price of oil
$NX_t \equiv X_t - M_t$	p ^{rice} = Retail price of rice
Employment block	PB = Primary balance
$emp_t = f(Y_t)$	P ^Y = GDP deflator
Price block	r ^{bl} = Bank lending rate
$\Delta \log(CPI_t) =$	r ^{cb} = BSP policy rate
$f(\Delta \log(p_t^{oil}), \Delta \log(p_t^{rice}), \Delta \log(DD_t), \Delta \log(xr_t))$	r ^{dd} = Effective interest rate on domestic debt
$\pi_t \equiv 100 \left(\frac{CPI_t}{CPI_{t-4}} - 1 \right)$	r ^{df} = Effective interest rate on foreign debt
Monetary block	RES = Debt residual (nominal)
$r_t^{cb} = f(\pi_t^e - \pi_t^T)$	rr ^{bl} = Real bank lending rate
$r_t^{t91d} = f(r_t^{cb}, PB_t/Y_t, \pi_t^e)$	rr ^{t10y} = Real 10-year Treasury rate
$r_t^{t10y} = f(r_t^{t91d}, \pi_t^e)$	rr ^{t91d} = Real 91-day Treasury rate
$r_t^{bl} = f(r_t^{t10y}, \pi_t^e)$	r ^{t10y} = 10-year Treasury rate
$rr_t^{t91d} \equiv r_t^{t91d} - \pi_t$	r ^{t10yUS} = US 10-year Treasury rate
$rr_t^{t10y} \equiv r_t^{t10y} - \pi_t$	r ^{t91d} = 91-day Treasury rate
$rr_t^{bl} \equiv r_t^{bl} - \pi_t$	RV = Total revenues (nominal)
$xrr_t = xr_t \left(\frac{CPI_t^{US}}{CPI_t} \right)$	RV ^{NTX} = Non-tax revenues (nominal)
$r_t^{dd} = f(r_t^{t10y})$	RV ^{TX} = Tax revenues (nominal)
$r_t^{df} = f(r_t^{t10yUS}, DY_t)$	RV ^{TXBIR} = Internal tax revenues (nominal)
Fiscal block	RV ^{TXBOC} = Customs revenues (nominal)
$\log(RV_t^{TXBIR}) = f(\log(Y_t^N))$	X = exports
$\log(RV_t^{TXBOC}) = f(\log(P_t^Y M_t), \log(p_t^{oil}), \log(xr_t))$	XP = Total expenditure (nominal)
$\log(RV_t^{NTX}) = f(\log(Y_t^N))$	XP ^{IND} = Domestic interest payments (nominal)
$RV_t^{TX} \equiv RV_t^{TXBIR} + RV_t^{TXBOC}$	XP ^{INF} = Foreign interest payments (nominal)
$RV_t \equiv RV_t^{TX} + RV_t^{NTX}$	XP ^{INT} = Interest payments (nominal)
$\log(XP_t^{PR}) = f(DY_t)$	XP ^{PR} = Primary expenditure (nominal)
$XP_t^{INT} \equiv XP_t^{IND} + XP_t^{INF}$	xr = nominal peso-dollar exchange rate
$XP_t \equiv XP_t^{PR} + XP_t^{INT}$	xrr = real peso-dollar exchange rate
$XP_t^{IND} \equiv r_t^{dd} \times D_{t-1}^D$	Y = GDP
$XP_t^{INF} \equiv \left(\frac{xrr_t}{xrr_{t-1}} \right) r_t^{df} D_{t-1}^F$	YD = disposable income
$PB_t \equiv RV_t - XP_t^{PR}$	Y ^N = nominal GDP
$D_t \equiv XP_t^{INT} + D_{t-1}^D + \left(\frac{xrr_t}{xrr_{t-1}} \right) D_{t-1}^F - PB_t + RES_t$	Y ^{WORLD} = World GDP
$D_t^D \equiv \alpha_t D_t$	α = Share of domestic debt in total
$D_t^F \equiv (1 - \alpha_t) D_t$	π = inflation rate
	π^T = inflation target (midpoint)
	π_t^e = expected inflation rate

Source: Corpus and Debuque-Gonzales (2023, p. 3).

Similar to Debuque-Gonzales and Corpus (2023, 2024), we continue to adopt a stylized framework where output is determined from the demand side, as in earlier Keynes-based

models and some other small macroeconometric models (Hammersland and Tr   2014; Kasimati and Dawson 2009).

Debuque-Gonzales and Corpus (2023) introduced the following shocks: “a positive shock to government consumption; a positive shock to world oil prices; and a recession in the country’s major export partners” (p. 32). Meanwhile, Debuque-Gonzales and Corpus (2024) made an extension by also considering the following shocks: “a world oil price shock, an exchange rate shock, and a primary spending shock” (p. 19). We build on Debuque-Gonzales and Corpus (2023, 2024) by introducing impulse (temporary) shocks to the exogenous variables and examine the reaction of the endogenous variables relative to their baseline paths from the deterministic dynamic simulation. Thus, we introduce election spending shock in the domestic demand block by simultaneously altering the private consumption and government consumption equations (e.g., introduce shocks through the structural equations in Figure 2, and/or inflate the variables by x percent, where x can be 0.10, 0.15, and 0.20). With this, we assess the ability of the model, simulated as a complete system, to generate forecasts that are close to the actual data. Both in-sample and out-of-sample model evaluations were conducted.

3.3. Estimation

Behavioral equations were estimated¹¹ using the Autoregressive Distributed Lag (ARDL) method in Error Correction Model (ECM) form. Lag lengths were optimally selected using the Akaike Information Criterion (AIC) restricted to a maximum of 2 lags. Cointegration between level variables was tested using the bounds test approach developed by Pesaran et al. (2001). We chose specifications such that estimated coefficients of variables that enter the long-run equation display signs consistent with theory; variables with parameters that failed to conform with expectations based on either theory or intuition were relegated to the short-run equation or omitted altogether. In cases where the bounds test indicated the absence of cointegration, behavioral relationships were modeled as a short-run equation in first differences. Residual diagnostic checks testing for homoskedasticity, serial correlation, and normality were performed to ensure model adequacy. Appendix 1 shows the results of the stability tests.

Quarterly data spanning from the first quarter of 2002 to the fourth quarter of 2023 (2022Q1 to 2023Q4) were used in the model, covering a more extended period than that employed by Debuque-Gonzales and Corpus (2023, 2024) by an additional four years. Data were sourced from the CEIC Economic Database¹² and all series were seasonally adjusted using the X-13 routine prior to estimation to ensure comparability and remove regular seasonal effects. Appendix 2 shows the descriptive statistics. Results from Augmented Dickey–Fuller (ADF) tests indicated that most series were either integrated of order $I(1)$ or stationary at level $I(0)$, confirming their suitability for econometric modeling. Appendix 3 shows the results of stationarity tests.

The inclusion of the coronavirus (COVID-19) pandemic years (2020 to 2023) is justified on both methodological and economic grounds. While we recognize that these years represent atypical macroeconomic conditions, we retained them in the sample to ensure a comprehensive representation of the Philippine economy’s evolution over time. See the succeeding section for a detailed discussion on how the model empirically accounts for pandemic-specific shocks.

¹¹ EViews was used to solve the model, combining estimated behavioral equations and identities to obtain the dynamic numerical solution for simulation. The model was solved using the Broyden solution algorithm. For a description, see IHS Markit (2020, pp.1044 and 1324).

¹² Offers a global database with macroeconomic data from official sources, international institutions, and alternative data to help economists and investors track and analyze global trends. Website: <https://www.ceicdata.com/en>.

From an econometric standpoint, excluding these years would truncate the dataset and risk introducing bias into the long-run estimates by omitting a structural shock of unprecedented magnitude that influenced virtually all macroeconomic aggregates. Including this period allows the model to capture and differentiate extraordinary, non-cyclical disturbances from regular election-related fluctuations, thereby enhancing the robustness of parameter estimates and improving the model's out-of-sample forecasting accuracy.

Second, from an economic standpoint, the pandemic period provides valuable insights into how policy responses and economic behavior adjust under extreme shocks, including fiscal stimulus, mobility restrictions, and shifts in consumption and investment patterns. These dynamics mirror the shock-transmission mechanisms, both demand- and policy-driven, that are central to this study's examination of election-induced disturbances. Including pandemic data thus broadens the model's empirical base, enabling it to account for both exogenous global crises and domestic political cycles, and to test whether election shocks produce comparable macroeconomic effects or interact with crisis-related volatility.

Hence, the 2002 to 2023 data points provide a comprehensive temporal coverage that encapsulates multiple electoral cycles, episodes of macroeconomic expansion and contraction, and the unprecedented COVID-19 disruption, offering a richer foundation for evaluating the transmission of election shocks within the Philippine macroeconomy.

4. Results and discussion

4.1. Model evaluation

Following the earlier iterations by Debuque-Gonzales and Corpus (2023, 2024), we also focus on the model's dynamic forecasting performance. As indicated earlier, our sample period was extended to cover data from 2002Q1 to 2023Q4, thereby incorporating recent economic developments and policy episodes. This inclusion necessitated accounting for the COVID-19 pandemic period, an unprecedented disruption to the Philippine economy.

To control for pandemic effects, a dummy variable, equal to one from 2020Q2 to 2021Q2, was introduced in the behavioral equations. This period represents the span of maximum economic disruption and intensive policy response. The starting point coincides with the country's entry into a technical recession (Congressional Policy and Budget Research Department [CPBRD] 2020), while the endpoint reflects the onset of sequential economic recovery and the rollout of mass vaccination campaigns (Department of Health [DOH] 2021; Philippine Statistics Authority [PSA] 2021; World Bank [WB] 2021). While the use of a single binary variable simplifies a complex and evolving crisis, it provides a tractable way to capture the pandemic's immediate structural shock and the government's corresponding emergency policy interventions within the model's framework. Future refinements may consider alternative specifications, such as structural-break tests, regime-switching parameters, or distributed-lag formulations to better encapsulate pandemic-related uncertainties.

Model evaluation, as seen in Figure 2 and Table 2, focused on assessing predictive capability using standard forecast accuracy metrics, specifically, the mean absolute percentage error (MAPE) and mean absolute error (MAE). The MAPE was applied to level variables, while the MAE was used for percentage and rate variables, as seen in Table 2. Overall, forecasts for GDP and GDP growth demonstrated reasonable predictive accuracy, with estimated paths closely tracking the actual de-seasonalized series. Among GDP components, private consumption,

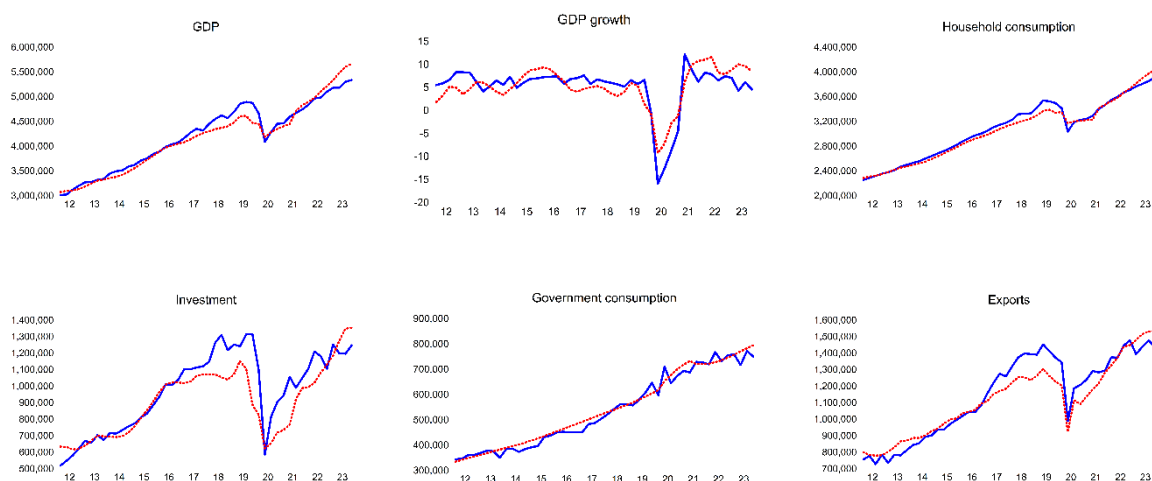
government consumption, and exports registered MAPEs of approximately 2%, 4%, and 5%, respectively. Slightly higher errors were observed for investment and imports, reflecting modest divergence between actual and predicted values during 2017 to 2022. This affected net exports toward the end of the sample.

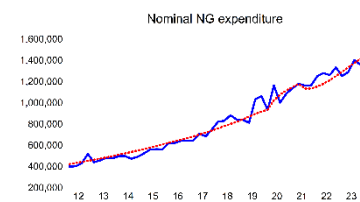
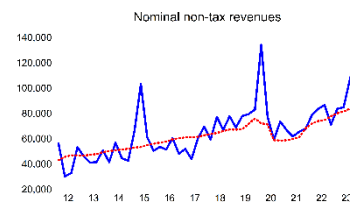
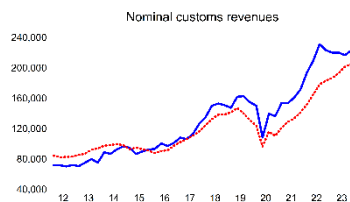
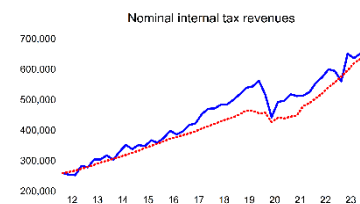
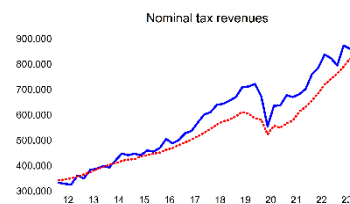
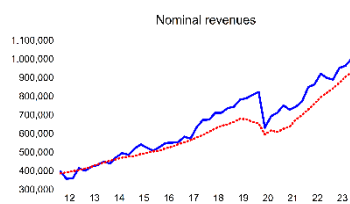
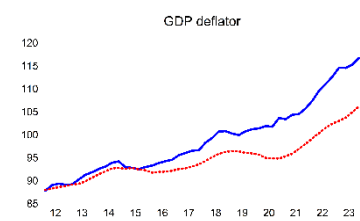
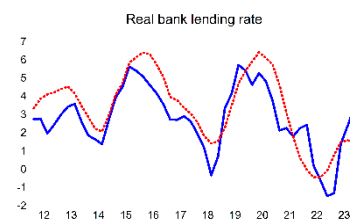
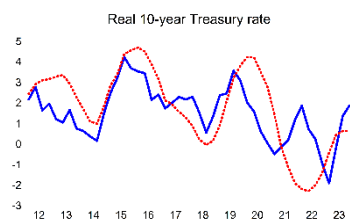
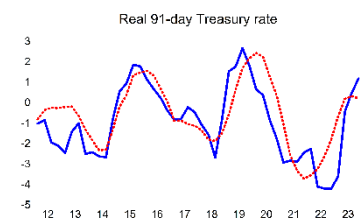
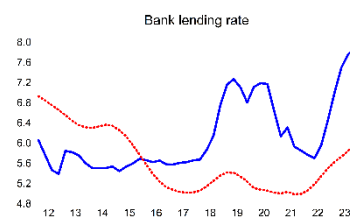
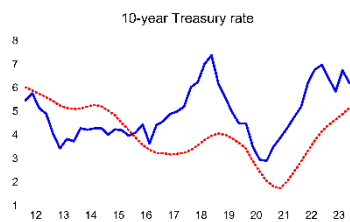
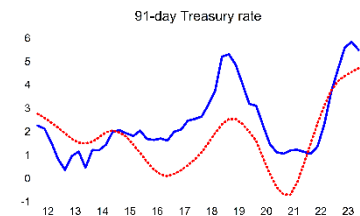
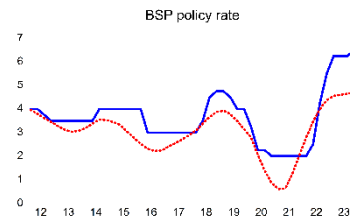
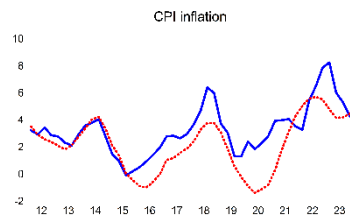
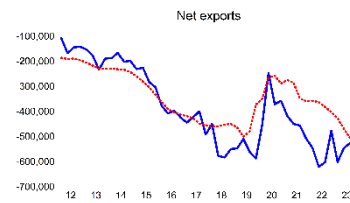
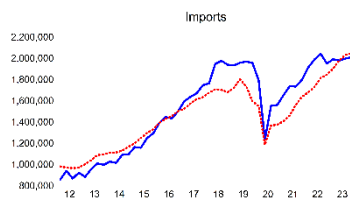
For tax revenues, including internal and customs revenues, the MAPE remained below 10%, only marginally higher than in earlier model versions. Non-tax revenues, however, recorded a higher MAPE of around 15%, due mainly to transient spikes during specific quarters, as seen in Figure 2. Interest payments followed the general trend of the actual data, though the forecast error was elevated in the first two years and again toward the latter part of the sample, particularly when disaggregated into domestic and foreign components. Dynamic forecasts for national government debt aligned closely with observed data, though a widening gap was noted near the end of the series. However, the MAPE was nearly identical from the previous iteration.

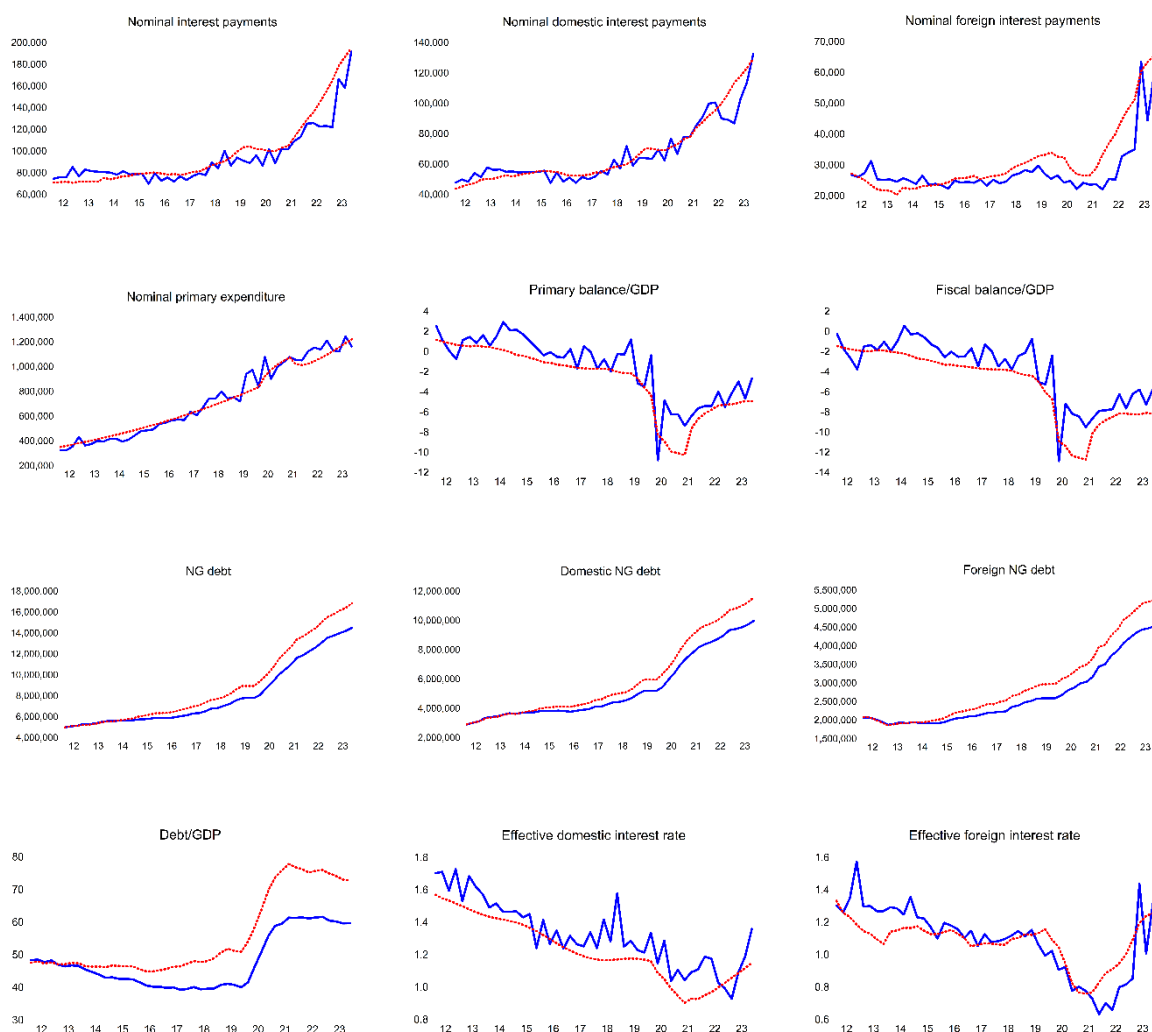
For percentage and rate variables, the model also displayed satisfactory predictive performance. Employment rate had an MAE of about 0.7, maintaining close alignment with observed trends. Inflation forecasts showed minor deviations, especially during 2020, consistent with pandemic-induced price volatility. Similarly, forecasts for bank lending and for 91-day and 10-year treasury bill rates exhibited modest divergences yet remained within acceptable margins, with MAEs of roughly 1.2, 2.3, and 2.0, respectively. When adjusted for real terms, these financial variables displayed forecast paths consistent with observed data.

The remaining forecast deviations, particularly during and immediately after the pandemic period, likely reflect the heightened uncertainty and nonlinear adjustments that are inherently challenging to capture within a small-scale macroeconomic model. Nonetheless, the extended model maintains a commendable level of accuracy, demonstrating its robustness in tracking both normal cyclical fluctuations and exceptional economic shocks such as the COVID-19 crisis.

Figure 2. In-sample simulations







Note: Blue lines are the actual data, while the broken red lines are the dynamic forecasts.

Source: Authors' calculations.

Table 2. Evaluation of in-sample forecast accuracy, 2021Q1-2023Q4

Mean absolute percentage error (MAPE) of level variables, in percent		
	GDP	2.617
	Household consumption	2.280
	Investment	9.694
	Government consumption	4.186
	Exports	5.223
	Imports	7.923
	Net exports	22.634
	Nominal revenues	6.236
	Nominal tax revenues	5.997
	Nominal internal tax revenues	6.687
	Nominal customs revenues	8.682
	Nominal non-tax revenues	15.820
	Nominal NG expenditure	5.728
	Nominal interest payments	19.242
	Nominal domestic interest payments	20.704

Nominal foreign interest payments	17.509
Nominal primary expenditure	6.533
NG debt	7.737
Domestic NG debt	7.737
Foreign NG debt	7.734
GDP deflator	4.386
Mean absolute error (MAE) of rate and percentage variables, in percentage points	
GDP growth	2.448
Employment rate	0.691
CPI inflation	2.293
BSP policy rate	1.315
91-day Treasury rate	2.287
10-year Treasury rate	1.961
Bank lending rate	1.212
Real 91-day Treasury rate	1.780
Real 10-year Treasury rate	1.749
Real bank lending rate	1.987
Effective domestic interest rate	0.197
Effective foreign interest rate	0.117
Revenue/GDP	0.677
Tax revenue/GDP	3.011
Internal tax revenue/GDP	0.328
Customs revenue/GDP	0.266
Non-tax revenue/GDP	0.267
Expenditure/GDP	1.240
Primary spending/GDP	1.428
Interest payments/GDP	0.379
Foreign interest payments/GDP	0.106
Domestic interest payments/GDP	0.292
Primary balance/GDP	1.418
Fiscal balance/GDP	1.286
Debt/GDP	3.805
Domestic debt/GDP	2.520
Foreign debt/GDP	1.287

Source: Authors' calculations.

4.2. Impact analysis of election-related spending shock

Defining election spending shock. We define an election spending shock as the four quarters preceding a Philippine national election, consistent with the empirical timing of political budget cycles observed across both developed and emerging economies (Brender and Drazen 2005; Shi and Svensson 2006; Drazen and Eslava 2010). In our study, shocks are introduced in 2015Q3 to 2016Q2 and 2018Q3 to 2019Q2, corresponding to the lead-up periods of the 2016 and 2019 elections.

Scope of macroeconomic variable selection. While Table 1 lists various macroeconomic variables used in the model, we only focus on private consumption, employment, investment, and government consumption because these variables represent the core transmission channels of election-related shocks identified in theory and empirical literature. That is, election cycles primarily operate through fiscal expansions and liquidity injections, influencing aggregate demand rather than supply-side fundamentals (Rogoff and Sibert 1988; Drazen and Eslava

2010). These four variables capture the direct behavioral responses of households, firms, and the public sector to election-driven spending.

Our selected macroeconomic variables also exhibit clear and measurable variations during election years in Philippine national accounts. In contrast, other macroeconomic indicators such as inflation, external balance, or interest rates tend to be indirectly affected, with weaker or lagged responses that are harder to isolate from concurrent global or policy shocks (e.g., commodity price movements, monetary tightening). Equally important, given the small-scale structure of the macroeconometric model, including additional variables risks introducing multicollinearity and parameter instability. Restricting interpretation to the principal demand-side variables ensures a more robust and interpretable simulation of election-shock transmission. Ultimately, our selected macroeconomic variables also provide the most actionable insights for fiscal and development planning. Understanding the dynamics of consumption, investment, employment, and public expenditure directly informs countercyclical policy design and resource allocation in election years.

Hence, by concentrating on these core aggregates, we capture the most salient and empirically verifiable pathways through which election shocks influence the Philippine economy, while maintaining methodological rigor and analytical focus.

Shock assumptions. To simulate these shocks, private domestic demand, comprising household consumption and private investment, is assumed to increase by 7%, 8%, 14%, and 9% in each of the four pre-election quarters, while government consumption rises by 8%, 9%, 15%, and 10%, respectively. These magnitudes are grounded in both domestic and international empirical evidence showing that election years are typically associated with elevated economic activity, primarily driven by heightened public expenditure and election-induced private spending. That is, the percentages introduced were based on the historical change in consumption and government spending for that particular quarter in the Philippines. Studies such as Rogoff and Sibert (1988) and Schultz (1995) established that incumbents often engage in expansionary fiscal policies prior to elections, while Akhmedov and Zhuravskaya (2004) and Drazen and Eslava (2010) confirmed that such spending creates short-lived surges in output and consumption. Philippine evidence also supports this pattern. Evangelista and Libre (2008) and Habito (2013) noted that national income growth typically accelerates in election years due to campaign-related activities, expanded government outlays, and higher liquidity in circulation.

Thus, our shock assumptions reflect the observed behavioral and fiscal regularities of the Philippine political cycle namely, frontloaded government spending, elevated household consumption, and temporary investment expansion driven by improved liquidity and sentiment. Consistent with the approach adopted in political business cycle literature, the model is solved under both baseline and shock scenarios for the sample period 2012Q1 to 2023Q4, generating dynamic forecasts for endogenous variables. These simulations, as seen in Figure 3, provide a quantitative basis to evaluate how pre-election demand impulses transmit through the Philippine macroeconomy and how quickly such effects dissipate once election-related activities subside.

Response of private consumption. Solving the model under the generated shock scenario produced a notable increase in private consumption relative to the baseline. During the 2016 election cycle, consumption rose by approximately 8.3%, 9.5%, 15.7%, and 10.7% in 2015Q3 to 2016Q2, while in the 2019 cycle it increased by 10.7%, 10.8%, 18.2%, and 14.4% in 2018Q3

to 2019Q2. These results confirm that election periods are associated with discernible surges in private domestic demand, consistent with the empirical evidence of pre-election fiscal expansions observed in both developing and emerging economies (Brender and Drazen 2005; Shi and Svensson 2006; Drazen and Eslava, 2010).

From Figure 3, the model also captures the transitory nature of this consumption boost. In the quarters immediately following an election, household consumption reverts toward its baseline trajectory, reflecting the dissipation of election-related spending once campaign activities conclude. The temporary spike in consumption highlights the unsustainable and short-lived nature of election-induced demand surges, a feature well-documented in studies of political budget cycles (Rogoff and Sibert 1988; Schultz 1995).

This short-term uplift is partly driven by policies and fiscal measures implemented or accelerated ahead of elections to increase disposable income, such as cash transfers, subsidies, or targeted social aid programs (Labonne 2016; Lokshin et al. 2022). These measures often expand liquidity among select demographic segments, temporarily boosting household consumption without creating lasting welfare gains. Additionally, political campaign spending, through wages, logistics, and advertising, injects short-term liquidity into the economy, stimulating short-lived consumer spending (Olano 2019).

Beyond these direct income effects, consumer sentiment also appears to play a role. As Cipullo and Reslow (2022) observed, governments tend to release optimistic growth forecasts before elections, potentially shaping public expectations and encouraging households to spend more in anticipation of continued economic expansion. However, such optimism often proves temporary, leading to a normalization of consumption behavior once post-election realities and fiscal adjustments set in.

Response of employment. Furthermore, the injection of liquidity into households during election periods can be partly attributed to campaign mobilization and the creation of temporary “election jobs.” As seen in Figure 3, employment rises in tandem with heightened economic activity, particularly in the first quarters of election years (2016Q1 and 2019Q1), when the model estimates an average employment increase of about 2.7 percent relative to the baseline. This result aligns with evidence from political business cycle studies showing that pre-election spending spurs temporary employment in construction, logistics, and service sectors associated with campaign and government activities (Akhmedov and Zhuravskaya 2004; Labonne 2016).

However, the model also reflects the ephemeral nature of such gains. Once election-related activities subside, the employment variable converges toward its baseline level, signifying the disappearance of short-term and informal election-linked jobs. This cyclical pattern mirrors findings in emerging economies, where politically induced labor demand expands briefly before elections and subsequently contracts when fiscal and campaign spending normalizes (Shi and Svensson 2006; Drazen and Eslava 2010).

Response of investment. For investment, the simulation indicates deviations of approximately 4.5%, 5.4%, 10.8%, and 9.1% from the baseline during 2015Q3 to 2016Q2, reflecting a discernible rise in capital formation during the pre-election period. From Figure 3, investment activity continues to strengthen modestly in the post-election quarters, suggesting an improvement in investor sentiment once political uncertainty subsides. The results imply that while pre-election fiscal expansion temporarily boosts demand-driven investment, the

subsequent resolution of electoral uncertainty can foster a mild but sustained rebound in private capital spending.

Empirical evidence supports this dual mechanism. Although findings on election-related investment behavior are mixed, there is broad consensus that election periods stimulate private domestic demand, creating short-term conditions favorable to investment (Le et al. 2024; Azzimonti 2024; Inosante 2025). Heightened consumption and government expenditure may prompt firms to expand production capacity or upgrade operations to meet increased demand. Moreover, sector-specific effects emerge as industries directly linked to electoral activity, such as media, advertising, logistics, and printing, register temporary surges in output, leading to targeted capital injections in these areas (Drazen and Eslava 2010).

At the same time, election cycles may influence investment indirectly through infrastructure continuity. The sustained implementation of Public-Private Partnership (PPP) projects, many of which are exempted from the election spending ban, provides a stabilizing channel for private investment even amid political transitions (Commission on Elections [COMELEC] 2018; Aning, 2024). This finding aligns with cross-country studies highlighting that policy credibility and ongoing public investment programs can partially offset uncertainty effects during election periods (Julio and Yook 2012; Shi and Svensson, 2006). Overall, the simulated outcomes suggest that while election shocks generate short-term investment accelerations, the extent to which these persist depends on the post-election policy environment and the continuity of major infrastructure initiatives.

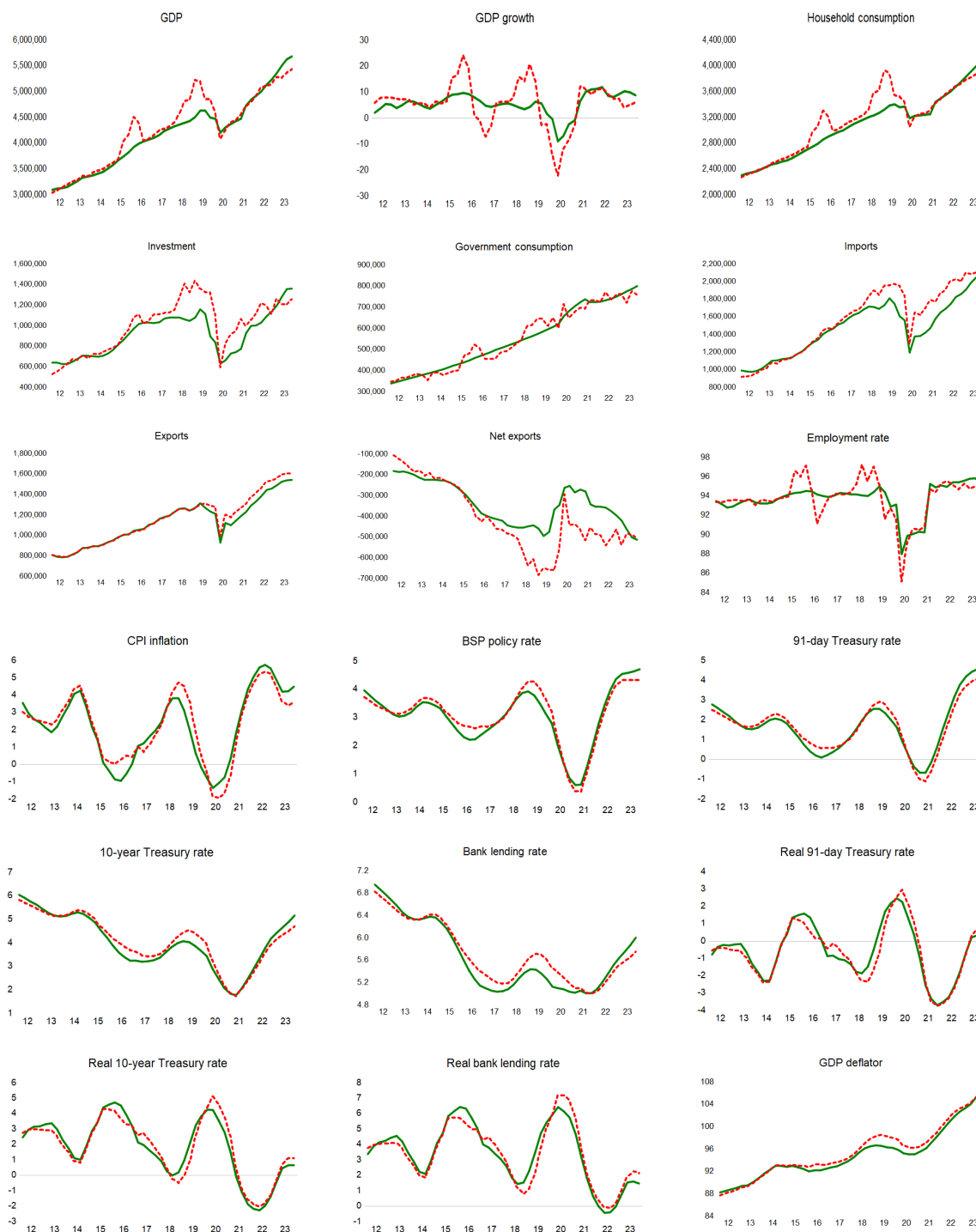
Response of government consumption. The shock analysis reveals that government consumption increased by approximately 7% to 14% in the quarters preceding the 2016 election and by 9.5% to 12.6% ahead of the 2019 election. This pattern affirms that Philippine election years typically witness an acceleration of public expenditure in anticipation of the statutory election spending ban, encompassing infrastructure outlays, social assistance programs, and the administrative costs of election operations (Das et al. 2025; Cigara 2025). Similar to household consumption, government spending displays a cyclical moderation in non-election years, returning to near-baseline levels once electoral activities conclude.

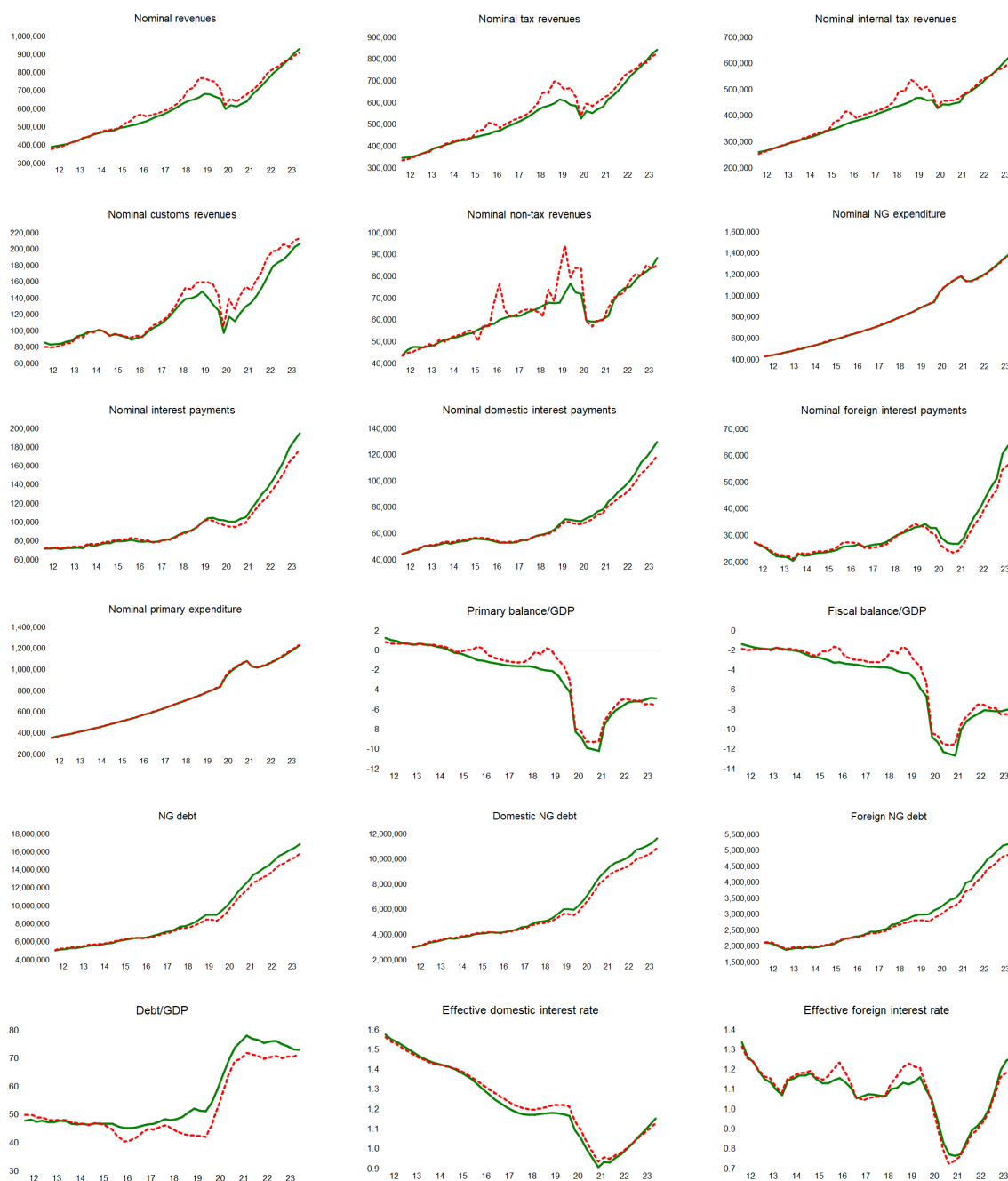
These findings align with the broader PBC literature, which demonstrates that incumbents often employ expansionary fiscal policies before elections to influence voter perceptions, followed by contractionary adjustments post-election (Rogoff and Sibert 1988; Schultz 1995; Peters 2010; Labonne 2016). The model likewise shows spending slowdowns before and after elections, with government consumption trending close to or even slightly below the baseline between 2015Q3 and 2016Q2, mirroring the empirical results of Evangelista and Libre (2008) for the Philippines. This suggests a pattern of fiscal frontloading, an intentional acceleration of expenditures before the COMELEC spending ban, particularly for infrastructure projects and social welfare programs, which are often justified as development priorities but may also serve political signaling purposes.

However, such behavior carries macroeconomic and developmental trade-offs. While election-driven fiscal expansions provide a temporary stimulus, they can also distort expenditure composition, diverting resources toward short-term, highly visible projects at the expense of long-term investments (Labonne 2016; De Haan et al. 2023; Punongbayan 2025). This reallocation may reduce fiscal efficiency, undermining the sustainability of growth once political incentives dissipate. Thus, results echo the consensus in empirical research. Although

election-induced spending cycles can generate brief economic gains, their cumulative effect may weaken fiscal discipline and development outcomes in the long run.

Figure 3. Election spending shock scenario





Note: Green lines are the baseline, while the broken red lines are the shock scenario.

Source: Authors' calculation.

Key findings. From Figure 3, we can construe that election years in the Philippines generate short-term, demand-driven expansions in key macroeconomic aggregates, consistent with PBC theory. We highlight that private consumption surged significantly by 8% to 16% during 2016 and 11% to 18% during 2019 election quarters. This is reflective of higher disposable income, intensified campaign activity, and optimistic expectations. This confirms the temporary consumption boost identified by Rogoff and Sibert (1988), Shi and Svensson (2006), and Evangelista and Libre (2008). For employment, we have seen it rise by roughly 2.7% in early election quarters, capturing the creation of short-term “election jobs.” However, this effect quickly dissipated post-election, consistent with Akhmedov and Zhuravskaya (2004) and Labonne (2016).

Meanwhile, private investment exhibited moderate pre-election growth of 4% to 11% driven by increased demand and the continuation of PPP projects, though partly offset by election-related uncertainty (Julio and Yook 2012; Drazen and Eslava 2010). Similarly, government consumption expanded by 7% to 14% before elections and normalized afterward, mirroring evidence of fiscal frontloading and subsequent retrenchment typical of political budget cycles (Brender and Drazen 2005; De Haan et al. 2023).

Together, these patterns illustrate that election shocks transmit primarily through the demand side of the economy, via household spending, employment, investment, and government outlays, producing short-term macroeconomic acceleration that is not sustained once electoral incentives subside.

Validation with scholarly literature. To assess the model’s consistency with empirical findings, we benchmarked our simulation results against established scholarly literature across four analytical dimensions: (1) magnitude and timing; (2) transmission channels; (3) persistence; and (4) investment ambiguity. Table 3 summarizes the validation of these dimensions against key studies.

Table 3. Validation of empirical results

Dimensions	Results	Reference
Magnitude and timing	The four-quarter, front-loaded shock matches canonical PBC timing with visible fiscal expansions in the year before elections.	Shi and Svensson (2006); Brender and Drazen (2005)
Transmission channels	Results underscore demand-side transmission from consumption to output, employment, and composition effects in public outlays.	Drazen and Eslava (2010)
Persistence	Rapid post-election mean reversion in consumption and employment is consistent with PBC models where the incentive is temporary and with evidence that expansions do not durably raise trend growth.	Brender and Drazen (2005); Rogoff and Sibert (1988)
Investment ambiguity	The model’s short-run investment bump coexists with scholarly literature’s uncertainty-induced deferral at the microeconomic level suggesting that macroeconomic demand can partially offset firm-level caution in election-exposed industries.	Julio and Yook (2012).

Source: Authors’ tabulation.

5. Ways forward

5.1. Conclusions

We have examined the transmission of election shocks in the Philippine economy through an augmented macroeconometric model developed by Debuque-Gonzales and Corpus (2023, 2024). Using quarterly data from 2002 to 2023, the model simulated the effects of pre-election demand surges on key macroeconomic aggregates namely private consumption, investment, employment, and government consumption. Our results confirmed that election periods in the Philippines are associated with short-term, demand-driven expansions consistent with the predictions of PBC theory. Private consumption, government spending, and employment all rise significantly in the quarters preceding elections, reflecting heightened fiscal disbursements,

campaign-related expenditures, and temporary labor creation. Investment also shows moderate gains, supported by buoyant demand and infrastructure continuity, though partially tempered by election-related uncertainty.

However, these effects are transitory. Post-election normalization occurs rapidly as fiscal activities contract and private confidence adjusts, highlighting the cyclical and unsustainable nature of election-induced growth. The model's findings align with scholarly literature (Rogoff and Sibert 1988; Brender and Drazen 2005; Drazen and Eslava 2010), which similarly identifies temporary expansions followed by corrective adjustments in fiscal behavior.

Overall, we have underscored that while election shocks provide short-term stimulus, they also reveal structural vulnerabilities in fiscal discipline, expenditure allocation, and policy continuity. Election-related economic activity boosts aggregate demand but may distort development priorities, exacerbate clientelism, and undermine long-term fiscal sustainability.

5.2. Policy recommendations

Building on our findings, we outline policy measures to manage the short-term economic fluctuations arising from election-related activities. While we have seen that election periods provide a temporary boost to consumption, investment, and employment, these gains are largely transitory and fiscally driven. To ensure that such cycles do not compromise fiscal sustainability or long-term development priorities, the following recommendations, seen in Table 4, focus on strengthening institutional discipline, improving expenditure quality, enhancing transparency, and fostering macroeconomic stability during and beyond election years.

Table 4. Policy recommendations

Policy recommendation	Implementation scheme
Institutionalize fiscal rules and pre-election spending discipline	To mitigate the volatility associated with election cycles, government agencies should strengthen adherence to fiscal rules that limit discretionary spending before elections. Establishing medium-term expenditure frameworks and transparent reporting of pre-election disbursements can reduce the scope for politically motivated fiscal expansion.
Prioritize capital-forming and inclusive expenditures	Rather than short-lived consumption-driven outlays, fiscal efforts during election years should prioritize capital-forming investments (i.e., infrastructure, human capital, and climate resilience) that yield longer-term productivity gains. Such reorientation minimizes waste and ensures that electoral cycles do not compromise developmental integrity.
Enhance transparency in budget execution and electoral financing	Commission on Audit (COA), COMELEC, and Department of Budget and Management (DBM) should collaborate on real-time monitoring of government spending patterns during election periods. Public disclosure of infrastructure disbursements, subsidy releases, and transfers would improve accountability and help deter opportunistic fiscal manipulation.
Strengthen automatic stabilizers and counter-cyclical policy tools	To cushion the economy from post-election slowdowns, the government should enhance automatic stabilizers (e.g., unemployment insurance, targeted social protection, progressive taxation). This would help sustain aggregate demand and maintain stability without resorting to politically driven fiscal impulses.

Maintain continuity of infrastructure and PPP programs	Continuity of PPP and flagship infrastructure projects should be insulated from political transitions to maintain investor confidence. Expanding the exemption of ongoing PPP projects from the election spending ban ensures policy predictability, supporting private investment even during periods of political uncertainty.
Improve data collection and political economy analysis	We highlight the need for higher-frequency and disaggregated data on government spending and campaign-related activity to deepen analysis of political business cycles. Integrating political economy variables into national forecasting frameworks (e.g., DEPDev, DBM, BSP) would allow for more realistic macroeconomic projections during election years.

Source: Authors' tabulation.

5.3. Limitations and areas for future studies

While we have offered valuable insights into the transmission of election shocks within the Philippine economy, several limitations emerged in the course of model development and simulation that future research may address.

First is on model stability and specification issues. As with any small-scale macroeconometric model, maintaining parameter stability across extended periods remains a challenge. The inclusion of multiple structural breaks, such as the 2008 Global Financial Crisis (GFC), the COVID-19 pandemic, and changes in fiscal regimes, may have introduced instability in long-run relationships among variables. Future studies should explore model re-specification or re-estimation using time-varying parameters, structural break tests, or Bayesian updating techniques to improve robustness over changing macroeconomic environments.

Second is on the impact of atypical years and structural shocks. Our analysis covered years characterized by atypical and extraordinary economic conditions, particularly the COVID-19 pandemic (i.e., 2020 to 2022). While we have employed dummy variables to account for these shocks, such a simplified treatment may not fully capture the depth and persistence of pandemic-induced disruptions, including behavioral shifts in consumption, investment, and labor market dynamics. Future studies could adopt nonlinear or regime-switching models to distinguish between ordinary cyclical shocks and extraordinary global crises.

Third is on post-COVID-19 structural changes. We recognize that the post-pandemic period introduced significant structural changes (e.g., digital transformation, altered household spending patterns, supply chain reconfigurations) that the current model, calibrated on pre-2020 relationships, may only partially reflect. Thus, a full rewriting or recalibration of the macroeconometric model may be warranted to capture these evolving structural parameters, particularly in labor productivity, consumption behavior, and fiscal multipliers.

Fourth, future research could extend the analysis by estimating election-shock impacts at the sectoral level, specifically across agriculture, industry, and services. Such disaggregation would provide deeper insight into how political cycles affect sector-specific dynamics, such as employment and output composition. However, this approach would require consistent and high-frequency sectoral time series data, and may necessitate the development of separate sectoral macroeconometric models rather than an aggregate one, given differences in structural behavior and data availability.

Fifth, future research may also extend the analysis to examine the qualitative influence of election outcomes, particularly the characteristics of winning candidates or administrations. Beyond fiscal behavior and macroeconomic trends, future models could incorporate leadership attributes (e.g., integrity, credibility, professional background, policy platforms, and governance orientation) as variables that may shape post-election economic performance. This approach would allow for an assessment of how political leadership quality affects investor confidence, fiscal prudence, and macroeconomic stability. However, such an extension would require developing quantifiable proxies for leadership traits and compiling corresponding post-election economic indicators, posing both methodological and data challenges that warrant dedicated empirical exploration.

Sixth, is on the practical versus theoretical application. While we aligned with econometric rigor, the model's usability for real-time decision-making remains limited relative to the needs of the private sector. Business and investment analysts often prioritize practical forecasting and scenario tools over structural rigor. Future extensions may consider developing simplified or hybrid models such as integrating macroeconomic foundations with business-cycle indicators to enhance practical relevance for private-sector users while maintaining analytical credibility.

Thus, these limitations point to promising directions for future studies. Refining the model's structure, enhancing its adaptability to non-traditional shocks, and bridging the gap between academic modeling and policy or market applications can help build a more resilient, context-sensitive macroeconomic framework for the Philippines that is capable of capturing both cyclical and structural dynamics in a rapidly evolving economic landscape.

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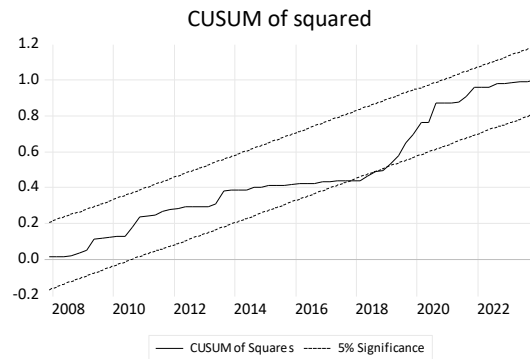
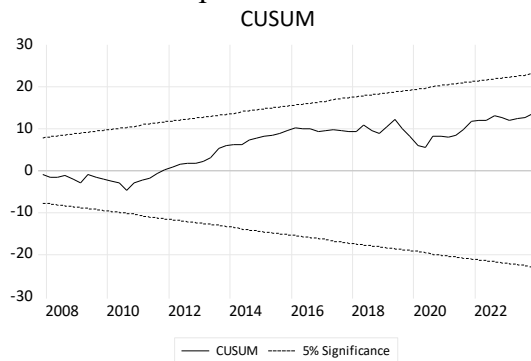
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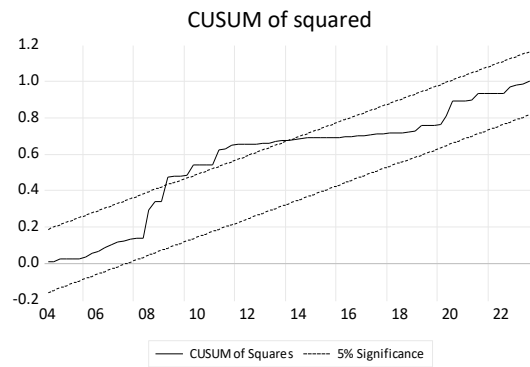
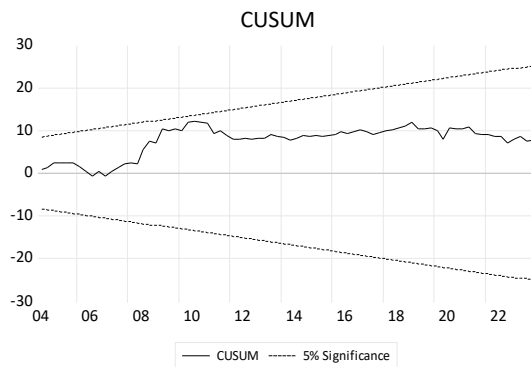
7. Appendix

Appendix 1. Cumulative sum (CUSUM) and CUSUM of squares test on equations

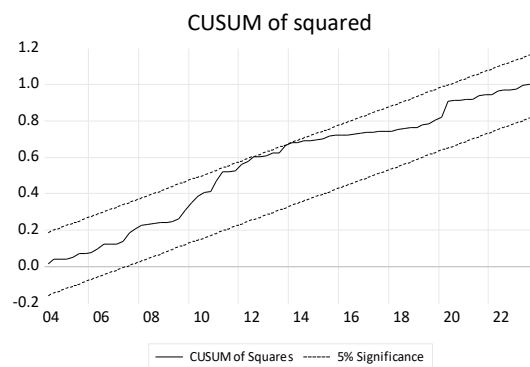
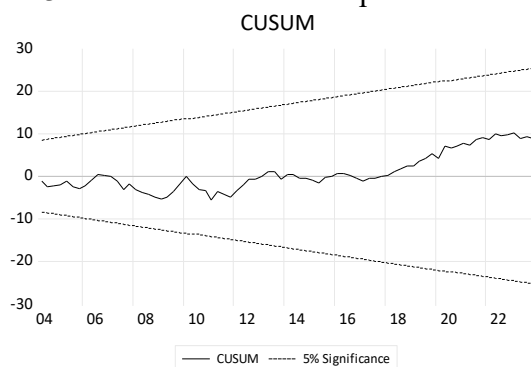
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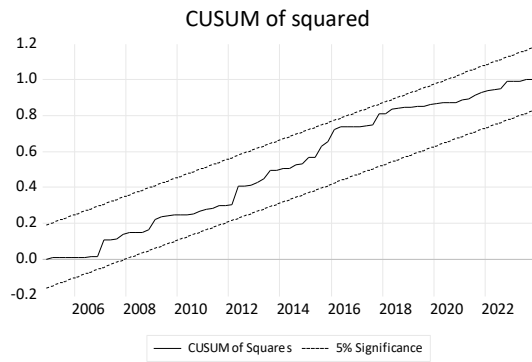
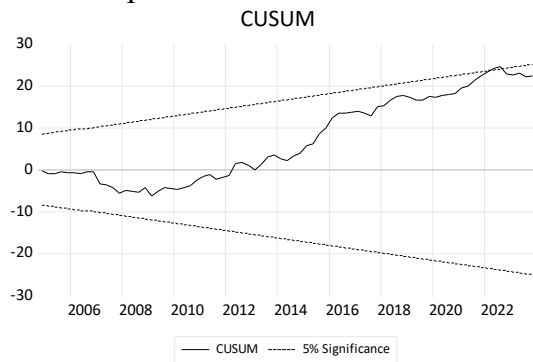
2. Investment



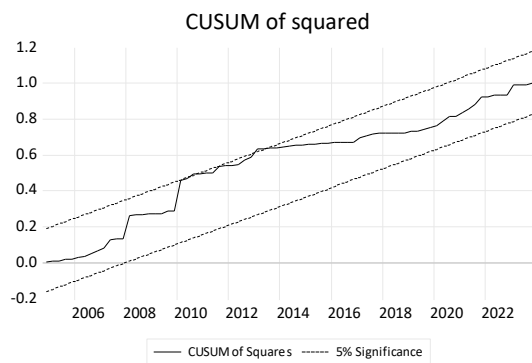
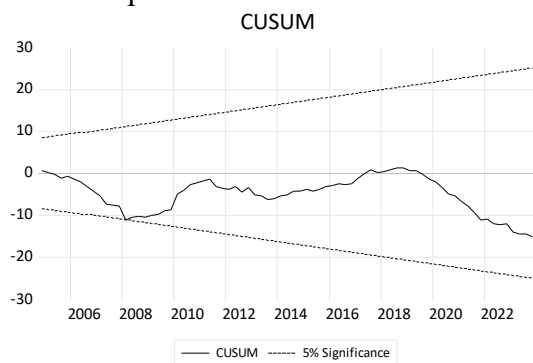
3. Government consumption



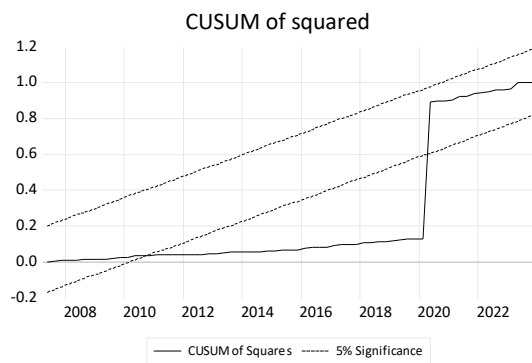
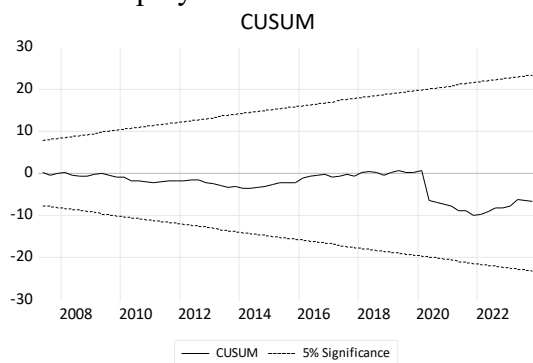
4. Imports



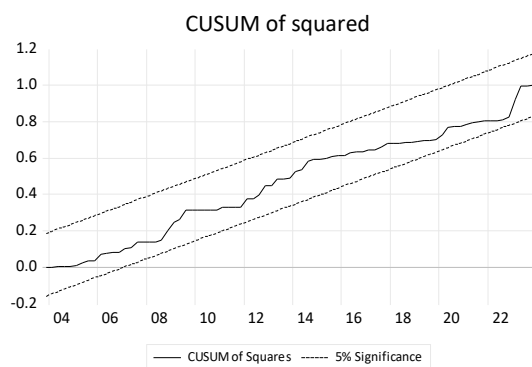
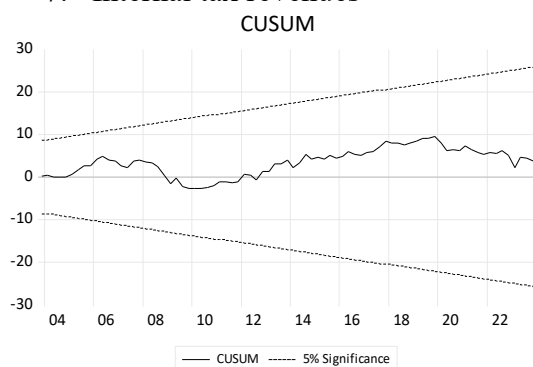
5. Exports



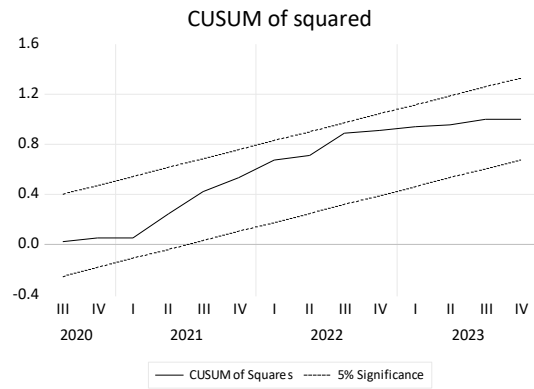
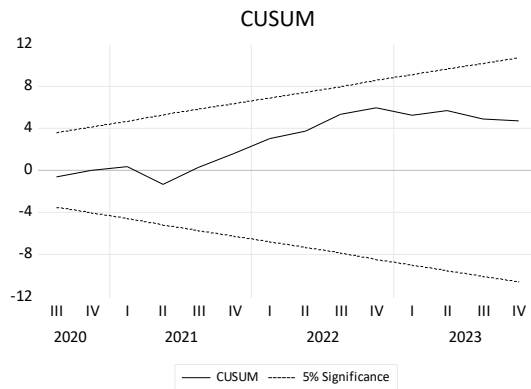
6. Employment rate



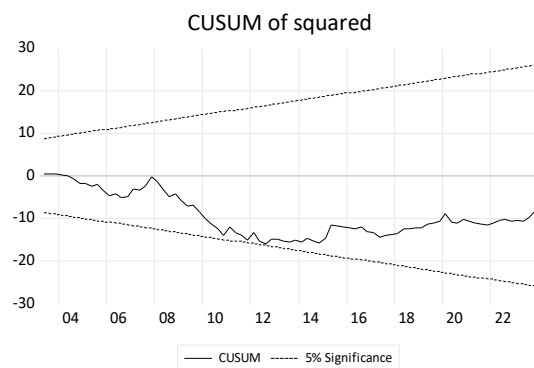
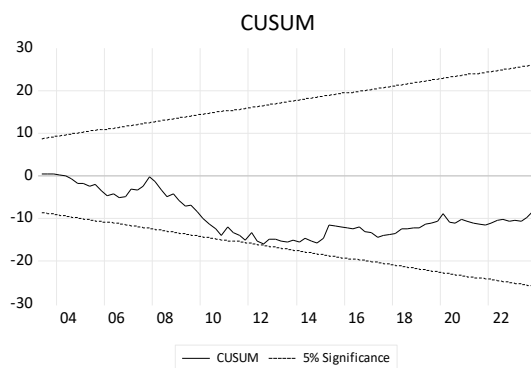
7. Internal tax revenues



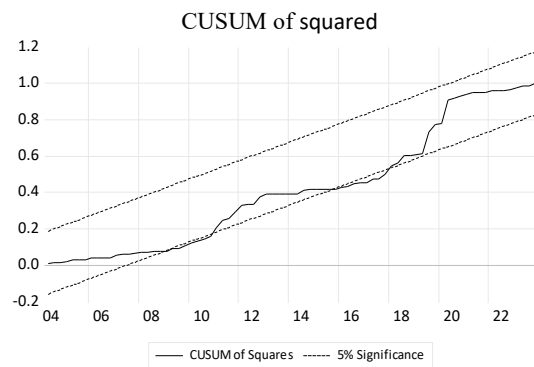
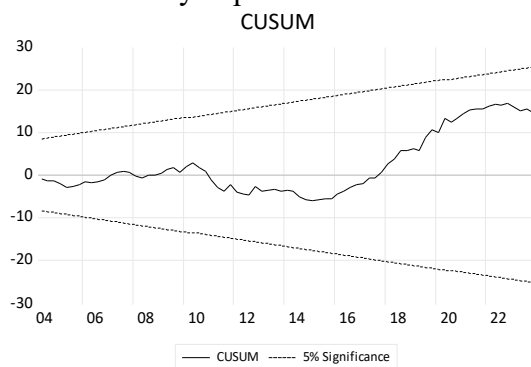
8. Customs revenues



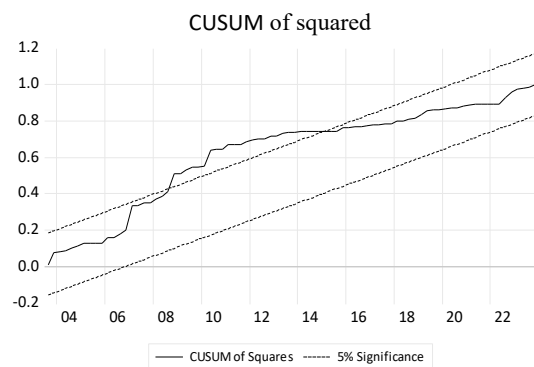
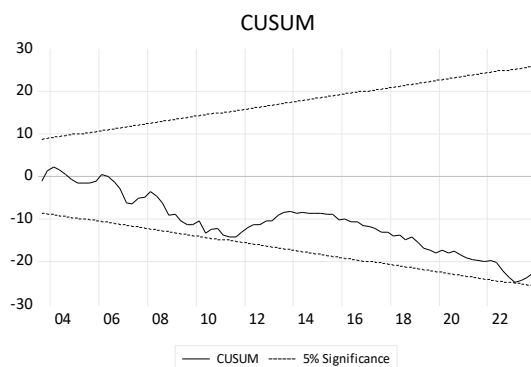
9. Non-tax revenues



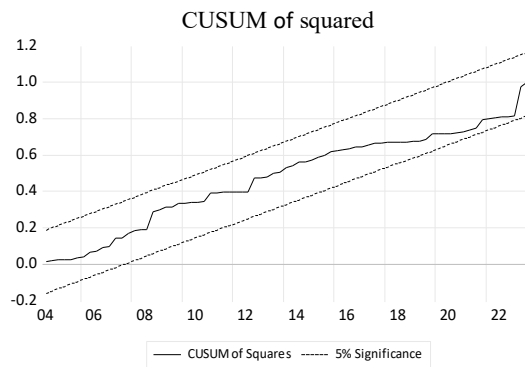
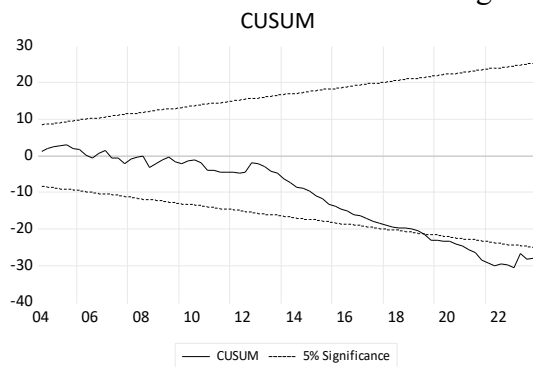
10. Primary expenditure



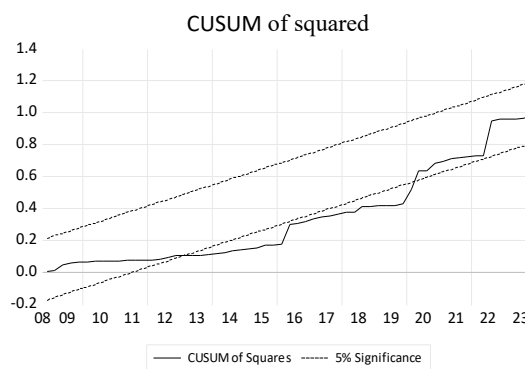
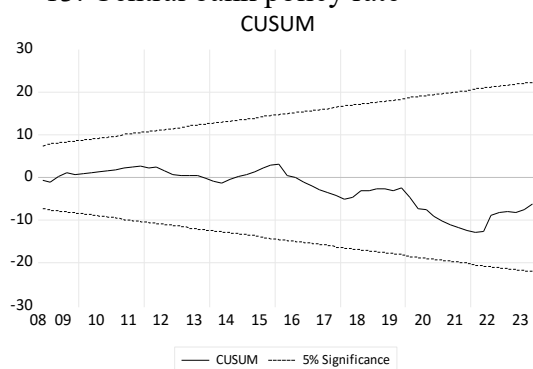
11. Effective interest rate on domestic debt



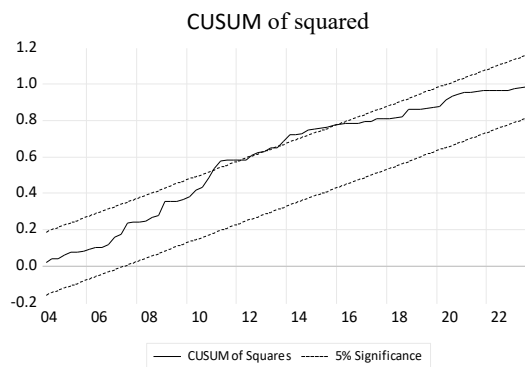
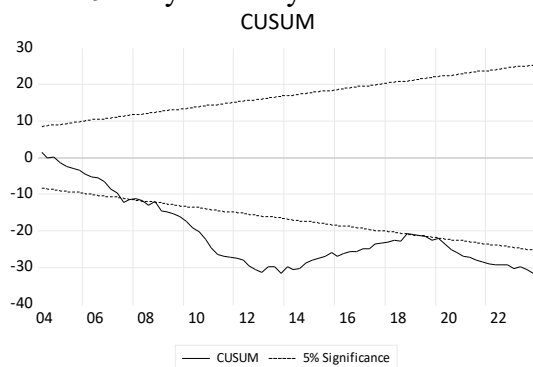
12. Effective interest rate on foreign debt



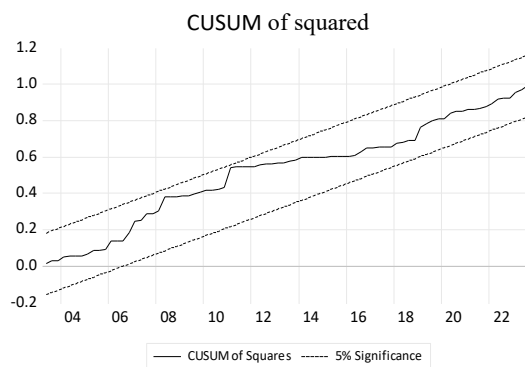
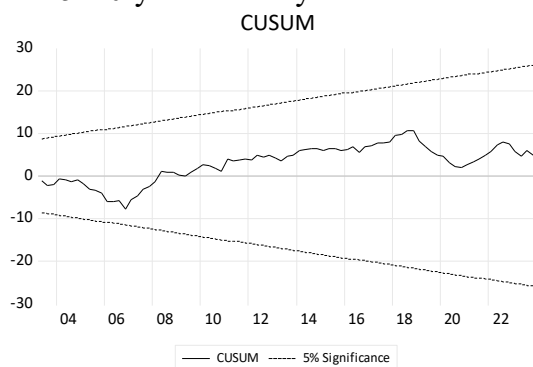
13. Central bank policy rate



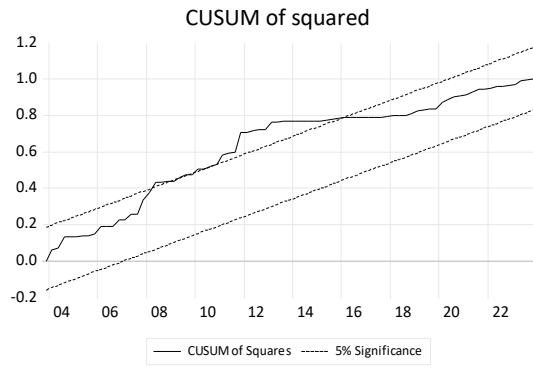
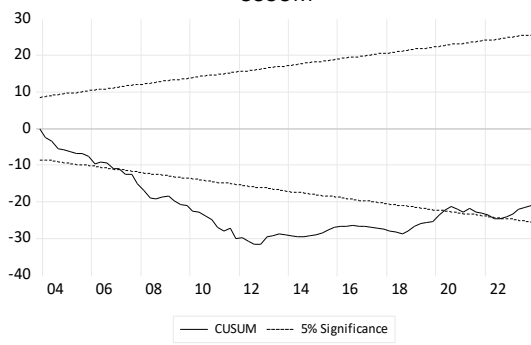
14. 91-day Treasury bill rate



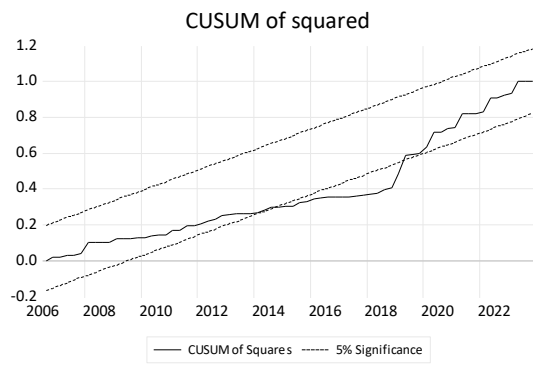
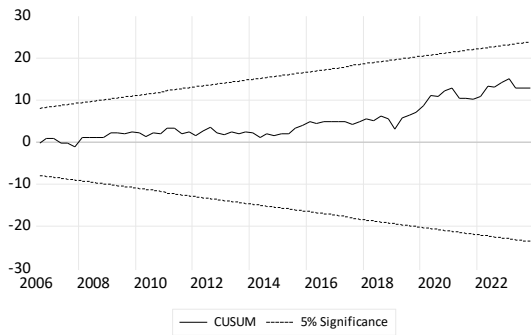
15. 10-year Treasury bond rate



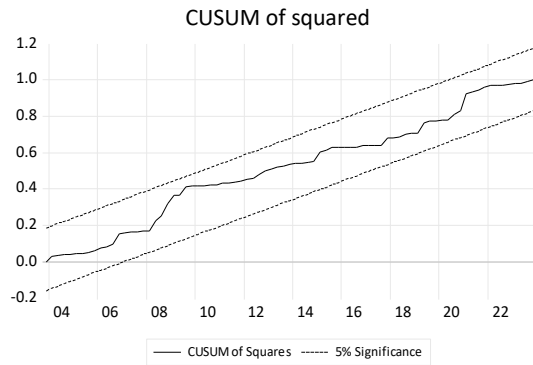
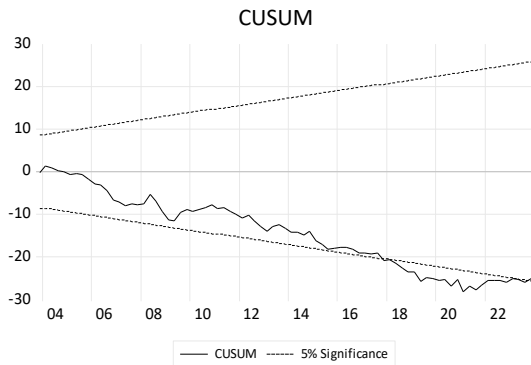
16. Bank lending rate



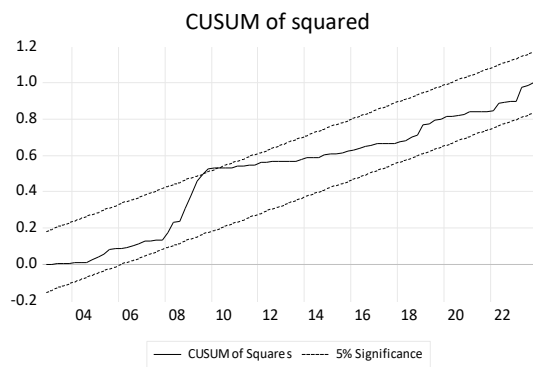
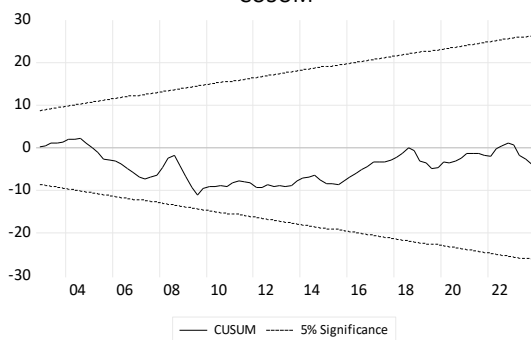
17. Consumer price index



18. GDP deflator



19. Inflation expectations



Appendix 2. Summary statistics

Variable	Obs	Mean	Std. dev.	Min	Max
GDP	92.00000	15.00616	0.337554	14.41938	15.54991
GDP growth	92.00000	5.100274	3.919953	-15.73532	12.28460
Household consumption	92.00000	14.71327	0.312916	14.16219	15.22696
Investment	92.00000	13.40527	0.493765	12.60449	14.12053
Government consumption	92.00000	12.85640	0.458427	12.17256	13.62048
Imports	92.00000	13.90079	0.442758	13.28778	14.56990
Exports	92.00000	13.67446	0.377367	12.99249	14.22927
Disposable income	92.00000	14.90120	0.330769	14.31227	15.43205
Domestic demand	92.00000	15.07274	0.360153	14.50052	15.64909
Employment rate	79.00000	93.32286	1.762076	82.68118	96.13102
Consumer price index	92.00000	4.434372	0.244569	3.988600	4.844825
GDP deflator	92.00000	4.462031	0.201113	4.048779	4.782431
US consumer price index	92.00000	4.584116	0.150793	4.320689	4.895367
CPI inflation	92.00000	3.863665	1.978919	-0.039411	10.31928
Deviation from inflation target	92.00000	0.081057	1.987872	-3.039411	6.319283
Expected inflation	90.00000	3.880644	1.826359	0.176666	10.17660
Deviation from inflation target	90.00000	0.102866	1.858691	-2.823334	6.176596
World oil price (USD per barrel)	92.00000	4.130562	0.440772	3.024053	4.767946
Retail price of ordinary rice (USD/ton)	81.00000	6.528090	0.233192	5.879447	6.814592
PHP/USD exchange rate	92.00000	3.896962	0.097006	3.706344	4.064011
Real PHP/USD exchange rate	92.00000	4.046706	0.146097	3.830282	4.358443
BSP policy rate	92.00000	4.785326	1.691357	2.000000	7.500000
91-day Treasury rate	92.00000	3.834284	2.138260	0.399254	8.133365
10-year Treasury rate	92.00000	7.014418	2.983422	2.933949	14.30097
Bank lending rate	91.00000	7.413242	1.642790	5.404006	10.85725
Real 91-day Treasury rate	92.00000	-0.029381	2.237194	-4.365562	5.032397
Real 10-year Treasury rate	92.00000	3.150752	2.947634	-1.868480	11.17548
Real bank lending rate	91.00000	3.535160	2.032507	-1.492220	7.816798
10-year Treasury rate US	92.00000	3.079301	1.144704	0.662028	5.069424
Nominal revenues	92.00000	12.94328	0.593574	11.81075	13.95539
Nominal tax revenues	92.00000	12.81778	0.605913	11.68525	13.80315
Nominal internal tax revenues	92.00000	12.55341	0.597581	11.42865	13.53053
Nominal customs revenues	92.00000	11.31126	0.647554	10.00319	12.37857
Nominal non-tax revenues	92.00000	10.76776	0.548466	9.662538	12.10431
Nominal NG expenditure	92.00000	13.13778	0.637177	12.16042	14.23665
Nominal primary expenditure	92.00000	12.94093	0.717014	11.84193	14.08569
Nominal interest payments	92.00000	11.31601	0.313734	10.58728	12.27139
Nominal domestic interest payments	92.00000	10.90898	0.362481	10.21010	11.83774
Nominal foreign interest payments	92.00000	10.19951	0.266079	9.429575	11.26906
Effective domestic interest rate	92.00000	1.635488	0.445163	0.931173	2.667550
Effective foreign interest rate	92.00000	1.267831	0.268807	0.636514	1.768446
Primary balance/GDP	92.00000	-0.282698	2.962809	-10.71427	4.635260
NG debt	92.00000	15.57367	0.481192	14.71613	16.59206
Domestic NG debt	92.00000	15.08119	0.575729	14.06369	16.20897
Foreign NG debt	92.00000	14.60770	0.339936	13.97840	15.45108
Debt/GDP	92.00000	53.31662	9.753641	39.46933	71.06895
Domestic debt/GDP	92.00000	32.47692	5.482577	25.62184	43.41447
Foreign debt/GDP	92.00000	20.83998	6.393255	13.38040	34.67981

Source: Authors' calculation

Appendix 3. Results of the ADF test

Variables	diff=0	diff=1	diff=2
GDP	0.848505	1.80E-08	2.79E-08
GDP growth	0.018645	2.22E-08	3.72E-08
Household consumption	0.850132	6.62E-08	0.000100
Investment	0.787998	0.000100	1.23E-07
Government consumption	0.973755	0.000100	0.000100
Imports	0.868088	0.000100	3.14E-08
Exports	0.719263	0.000100	0.000100
Disposable income	0.812715	8.59E-08	2.92E-08
Domestic demand	0.917530	4.13E-08	0.000100
Employment rate	0.000100	0.000100	0.000100
Consumer price index	0.733661	5.62E-05	2.38E-08
GDP deflator	0.364247	3.00E-07	3.50E-08
US consumer price index	0.991907	1.13E-06	2.53E-08
CPI inflation	0.007652	2.35E-05	2.13E-05
Deviation from inflation target	0.004059	0.001588	7.13E-08
Expected inflation	0.184991	3.38E-05	1.19E-07
Deviation from inflation target	0.004307	1.07E-06	1.40E-07
World oil price (USD per barrel)	0.071016	2.01E-08	2.35E-08
Retail price of ordinary rice (USD/ton)	0.059567	1.43E-07	0.000100
PHP/USD exchange rate	0.668310	4.04E-07	0.000100
Real PHP/USD exchange rate	0.510377	2.95E-07	5.39E-06
BSP policy rate	0.163745	2.44E-05	2.37E-08
91-day Treasury rate	0.031014	2.12E-06	0.000100
10-year Treasury rate	0.017979	1.10E-08	1.42E-05
Bank lending rate	0.014352	5.99E-08	1.22E-06
Real 91-day Treasury rate	0.015754	0.001393	3.10E-08
Real 10-year Treasury rate	0.008301	4.83E-08	5.39E-08
Real bank lending rate	0.001256	0.000383	6.18E-07
10-year Treasury rate US	0.278236	2.41E-08	1.56E-08
Nominal revenues	0.827929	2.09E-07	0.000100
Nominal tax revenues	0.839660	0.000100	3.34E-06
Nominal internal tax revenues	0.748273	0.000100	5.02E-08
Nominal customs revenues	0.683355	0.000100	0.000100
Nominal non-tax revenues	0.880388	0.000100	3.07E-08
Nominal NG expenditure	0.984335	0.000100	4.79E-07
Nominal primary expenditure	0.968874	0.000100	1.52E-06
Nominal interest payments	0.988705	0.000100	4.16E-08
Nominal domestic interest payments	0.969265	0.000100	4.36E-08
Nominal foreign interest payments	0.873016	0.000100	3.33E-05
Effective domestic interest rate	0.470250	0.000100	3.45E-06
Effective foreign interest rate	0.401333	2.35E-08	0.000100
Primary balance/GDP	0.358776	1.08E-05	1.52E-06
NG debt	0.996948	6.83E-05	8.54E-07
Domestic NG debt	0.989639	8.17E-06	0.000100
Foreign NG debt	0.998088	1.00E-06	7.32E-05
Debt/GDP	0.481812	0.002019	0.000100
Domestic debt/GDP	0.548330	0.000683	0.000100
Foreign debt/GDP	0.601342	0.047112	0.000100

Source: Authors' calculation